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Enhanced particle swarm optimization for feature selection in SVM-based Alzheimer's disease diagnosis

Qian Zhang[1,2,3], Jinhua Sheng[1,2,9]✉, Rougang Zhou[2,4,5], Qiao Zhang[6,7,8], Binbing Wang[1,2] & Rong Zhang[1,2]

Alzheimer's Disease (AD) is a progressive neurodegenerative disorder marked by neuronal loss, leading to cognitive and behavioral decline. With the aging global population, AD incidence and its socioeconomic burden are increasing. Developing effective early diagnostic methods is thus critical for improving patient outcomes and slowing disease progression. In this paper, an enhanced Particle Swarm Optimization (PSO) algorithm, which integrates opposition-based Latin squares sampling initialization (OL) with dynamic inertia weights and learning factors (D), termed OLDPSO, is proposed to improve feature selection and classification within a Support Vector Machine (SVM) model for AD diagnosis using magnetic resonance imaging (MRI) data. MRI, as a non-invasive modality, reveals structural brain changes, particularly in gray matter (GM) and white matter (WM) volumes, which are key biomarkers for AD. However, extracting essential features from complex GM and WM data remains a significant challenge. To address this, the proposed OLDPSO, which adaptively balances global exploration and local exploitation, overcomes traditional PSO limitations. Benchmark experiments show that OLDPSO outperforms existing PSO variants in solution quality and convergence speed. Validated with data from the AD Neuroimaging Initiative (ADNI), the OLDPSO-SVM model demonstrates superior performance in differentiating AD, mild cognitive impairment (MCI), and normal control (NC) groups, particularly in classifying MCI subtypes (MCI-NC and MCI-C). Results show that combining GM and WM features yields higher diagnostic accuracy than using either alone, and the model identified key brain regions associated with AD progression. Specifically, the model achieved accuracies of 99.11%, 89.76%, 99.07%, 88.38%, 94.69%, and 87.96% in the diagnosis of AD vs. NC, NC vs. MCI-NC, NC vs. MCI-C, MCI-NC vs. MCI-C, MCI-NC vs. AD, and MCI-C vs. AD, respectively. Through optimized feature selection, the OLDPSO-SVM model enhances diagnostic performance and provides valuable insights for developing MRI-based multimodal diagnostic tools for AD.

Keywords Alzheimer's disease, Particle swarm optimization, Support vector machine, Feature selection, Gray matter, White matter

Alzheimer's Disease (AD)[1] is a progressive neurodegenerative disorder characterized by the gradual loss of neurons in the brain, leading to memory impairment, cognitive dysfunction, and behavioral changes. With the global aging population on the rise, the incidence of AD continues to rise, placing a substantial burden on patients, their families, and the economy[2–4]. Therefore, developing effective early diagnostic methods for AD is crucial for improving patient outcomes and delaying disease progression.

[1]School of Computer Science and Technology, Hangzhou Dianzi University, Hangzhou 310018, Zhejiang, China. [2]Key Laboratory of Intelligent Image Analysis for Sensory and Cognitive Health, Ministry of Industry and Information Technology of P. R. China, Hangzhou 310018, Zhejiang, China. [3]School of Data Science and Artificial Intelligence, Wenzhou University of Technology, Wenzhou 325035, Zhejiang, China. [4]College of Mechanical Engineering, Hangzhou Dianzi University, Hangzhou 310018, Zhejiang, China. [5]Mstar Technologies Inc, Hangzhou 310018, Zhejiang, China. [6]Beijing Hospital, Beijing 100730, China. [7]National Center of Gerontology, Beijing 100730, China. [8]Institute of Geriatric Medicine, Chinese Academy of Medical Sciences, Beijing 100730, China. [9]Institute of Cognitive and Intelligent Computing, Hangzhou Dianzi University, Xiasha Higher Education Zone, Hangzhou 310018, China. ✉email: j.sheng@ieee.org

In early AD diagnosis, magnetic resonance imaging (MRI)[5,6] provides a non-invasive means to observe structural changes in the brain. Specifically, analyzing volumetric changes in gray matter (GM) and white matter (WM) provides essential biomarkers for diagnosing AD[7,8]. Studies have shown that abnormal GM and WM alterations play critical roles in the pathogenesis of AD, with GM atrophy and WM damage closely linked to cognitive decline[9,10]. However, extracting key features from complex GM and WM data remains a significant challenge in AD diagnosis.

Mild Cognitive Impairment (MCI)[11,12] is considered an early stage of AD, characterized by a slight decline in cognitive function, which places individuals at a higher risk of progressing to AD. MCI can be further divided into two subtypes: non-converting MCI (MCI-NC), in which cognitive function remains relatively stable, and converting MCI (MCI-C), in which patients transition to AD within a shorter timeframe[13]. Since the imaging differences between MCI-NC and MCI-C are subtle, accurately distinguishing these subtypes remains a challenging diagnostic issue. Differentiating these MCI subtypes is essential for early AD diagnosis, as it not only aids in creating personalized treatment plans but may also help slow disease progression.

Support Vector Machine (SVM) has been recognized as a powerful classification algorithm and has been widely applied across various biomedical data analyses. In the realm of human activity recognition (HAR), SVM has been effectively used for health-related applications such as elderly care and lifestyle monitoring, enabling the classification of activities based on features extracted from smart device data[14]. Similarly, in driver drowsiness detection, SVM has been applied to electroencephalography (EEG) and blink-related features, improving classification accuracy by integrating blink feature extraction and artifact removal[15]. In the domain of medical diagnostics, SVM has shown great promise. For instance, SVM has been utilized to analyze plantar pressure data from active-matrix sensing arrays (AMSA) to accurately diagnose lumbar degenerative disease (LDD) and assess post-surgery recovery, achieving 100% accuracy in clinical tests[16]. Furthermore, SVM has demonstrated high performance in the classification of skin cancer through the analysis of dermoscopic images, achieving high accuracy in detecting malignancies across various datasets[17]. In neurodegenerative disease diagnosis, SVM has been applied to the quantitative analysis of phosphorylated tau proteins (p-tau) in plasma using a plasmonic asymmetric nanobridge (PAN)-based biosensor, distinguishing between normal control (NC), MCI, and AD with a diagnostic accuracy of 94.47%[18]. SVM has also been combined with pre-trained convolutional neural networks (CNNs) to address challenges such as data imbalance, as demonstrated in the classification of 19 white blood cell (WBC) types, where this combination achieved an accuracy of 95%[19]. Additionally, SVM has been applied to the analysis of extracellular vesicle (EV) data from plasma samples using surface-enhanced Raman spectroscopy (SERS), distinguishing between healthy, benign, and high-grade serous carcinoma (HGSC) cases with high precision[20]. By finding a hyperplane that maximizes the margin between classes, SVM has strong generalization capabilities, especially suitable for high-dimensional data classification. However, SVM performance heavily relies on the quality of the input features and model parameter optimization. When handling high-dimensional MRI data, effectively selecting the most discriminative features from extensive GM and WM volume characteristics and optimizing the SVM model parameters are essential for improving classification performance.

Particle Swarm Optimization (PSO) algorithm[21], as a well-established swarm intelligence optimization technique, has gained widespread application in machine learning tasks due to its notable search efficiency and ease of implementation. By imitating the foraging behavior of bird swarms, PSO iteratively searches for optimal solutions within the solution space, demonstrating strong global search capabilities. In recent years, PSO has also garnered increasing attention in medical diagnostics due to its distinct search mechanism and superior global optimization performance. For example, Zablocki et al.[22] applied the PSO algorithm to address issues related to centering and recurrence in optimally arranged biological sequences, a critical challenge in bioinformatics. Furthermore, Abdi et al.[23] utilized PSO to construct a novel weighted SVM, improving the categorization of samples in gene selection and tumor classification tasks. Wang et al.[24] applied PSO to a Self-Tuning Convolutional Neural Network (PSTCNN) for Coronavirus Disease 2019 (COVID-19) diagnosis, optimizing hyperparameters automatically and significantly improving diagnostic accuracy and efficiency. Similarly, Kaur et al.[25] applied PSO to optimize a CNN for cervical cancer diagnosis, enhancing model accuracy and performance across benchmark datasets. Manimurugan et al.[26] integrated PSO with a stacked autoencoder (SAE) for breast cancer classification in mammograms, achieving high accuracy and enhancing diagnostic precision. Alshraideh et al.[27] used PSO for feature selection, achieving 94.3% accuracy with an SVM classifier for heart disease prediction, thereby enhancing early detection and diagnosis. Aguerchi et al.[28] also applied PSO to optimize a CNN model for breast cancer detection. Nevertheless, despite these successes, traditional PSO encounters difficulties in high-dimensional problems, particularly in the later stages of optimization when particle diversity tends to decrease. This often leads to the algorithm becoming trapped in local optima, which subsequently reduces search efficiency and slows convergence. Therefore, enhancing the global search capabilities of PSO to mitigate these limitations remains a critical area of ongoing research.

To address the limitations of traditional PSO discussed earlier, this paper proposes an improved PSO algorithm (OLDPSO) that significantly enhances global search capability and convergence speed through a combination of opposition-based Latin squares sampling initialization and dynamic inertia weights and learning factors. First, opposite learning expands the search space by generating opposite solutions for the current particle positions, reducing the risk of local optima by increasing global exploration opportunities. Additionally, Latin squares sampling is a stratified random sampling technique that enhances the initial solution's uniformity and diversity, accelerating convergence. Furthermore, this study incorporates an adaptive adjustment mechanism for dynamic inertia weights and learning factors, allowing the particle swarm to flexibly balance global exploration and local exploitation at different optimization stages, further improving optimization efficiency. The effectiveness of the OLDPSO algorithm is validated through a series of experiments. The results indicate that OLDPSO exhibits superior performance on various benchmark functions compared to multiple existing PSO variants,

with significant improvements in solution quality and convergence speed. Particularly in handling complex optimization problems, OLDPSO effectively maintains particle diversity and prevents premature convergence to local optima. These experimental findings demonstrate the potential and efficacy of OLDPSO in solving complex optimization problems.

The primary objective of this study is to enhance the feature selection and classification performance of the SVM model by leveraging MRI-based GM and WM features through the optimized OLDPSO algorithm. As crucial components of brain structure, GM and WM exhibit distinct structural changes in the AD pathogenesis process, and using only one feature may not sufficiently reflect the full range of pathological characteristics. Therefore, this study combines GM and WM volumetric features to provide a more comprehensive and accurate feature representation for the classification and diagnosis of AD and its early stages, such as MCI. To achieve this, the OLDPSO algorithm is employed to optimize feature selection, ensuring the most discriminative GM and WM volumetric features are selected. Additionally, the SVM parameters are dynamically adjusted to improve the model's generalization and diagnostic performance. Experimental results demonstrate that the OLDPSO-SVM model significantly enhances performance in AD-related classification tasks, showing improved accuracy and stability compared to models using only GM or WM features. By optimizing both feature selection and classification models, this study offers an effective solution for the early diagnosis of AD.

The key contributions of this paper are as follows:

- This paper proposes a novel optimization algorithm, OLDPSO, which integrates opposition-based Latin squares sampling initialization with dynamic inertia weights and learning factors to enhance the global search capability and convergence speed of the traditional PSO algorithm, outperforming other enhanced PSO variants.
- The paper compares different transfer functions (TFs) and identifies the optimal TF for each AD classification task, further improving the classification accuracy and stability of the OLDPSO-SVM model in AD diagnosis.
- The study applies the optimized OLDPSO algorithm for feature selection and classification in AD diagnosis using MRI-based GM and WM features, significantly enhancing accuracy and robustness compared to models using only GM or WM.
- By leveraging the OLDPSO algorithm, the study identifies the most discriminative GM and WM volumetric features, providing a more comprehensive representation of brain structure for early detection of AD and its stages, such as MCI, including the challenging task of distinguishing between MCI-C and MCI-NC.

The structure of this paper is as follows: Sect. 2 presents the background of SVM and PSO algorithms. Section 3 elaborates on the proposed OLDPSO algorithm and its theoretical foundation. Section 4 describes data collection, preprocessing, feature extraction, and the construction of the OLDPSO-SVM model. Section 5 presents the experimental results of the OLDPSO algorithm. Section 6 applies the OLDPSO-SVM model to AD diagnosis and displays results. Section 7 provides an in-depth discussion of the experimental findings, and Sect. 8 concludes the study.

Background

An overview of SVM

SVM is a supervised machine learning algorithm that is widely used in classification and regression problems. Figure 1 shows the schematic diagram of SVM. Its core idea is to input a sample set $S = \{(x_n, y_n); x_n \in R^l; y_n \in R; n = 1, 2, \cdots, N\}$, where x_n is the input variable represented as an l-dimensional vector, y_n is the output variable, and N is the total number of samples in the dataset. Map it to

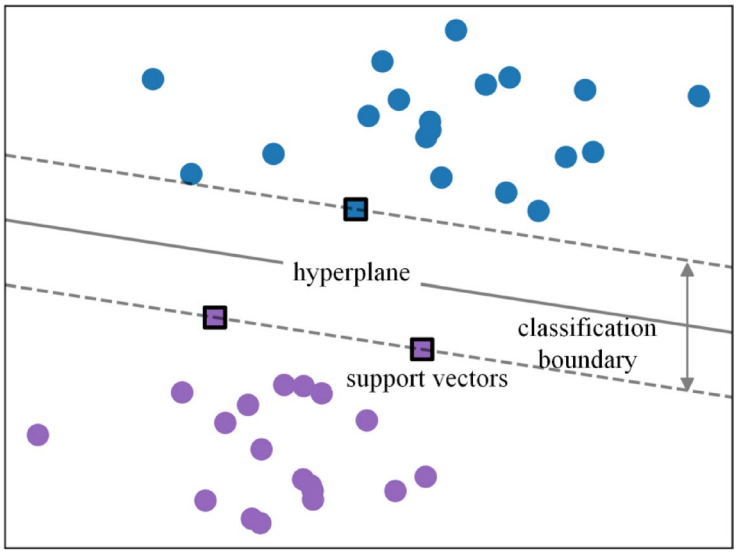


Fig. 1. Schematic of SVM.

a high-dimensional feature space and construct an optimal hyperplane, which can be regarded as a decision boundary in this space. The optimal hyperplane is used to split the sample set effectively, and support vectors are found on both sides of the hyperplane. These support vectors are the sample points closest to the hyperplane, with good generalization ability and robustness[29]. The optimal hyperplane for partitioning can be expressed following.

$$w^T x + b_s = 0 \tag{1}$$

where w is the optimal hyperplane normal vector and b_s is the distance between the hyperplane and the distant point. To find the most suitable b_s and w, we need to solve this optimization problem.

$$\begin{cases} min \frac{1}{2} ||w||^2 + C \sum_{n=1}^{N} (\xi_n + \xi_n^*) \\ s.t. \begin{cases} y_n - w^T x - b_s \leq \varepsilon + \xi_n \\ w^T x + b_s - y_n \leq \varepsilon + \xi_n^* \\ n = 1, 2, \cdots, N \end{cases} \end{cases} \tag{2}$$

where C is the penalty factor, ϵ is the insensitivity factor, and ξ_n and ξ_n^* are the relaxation factors.

To solve some sample sets that cannot be divided in linear space, the sample is mapped from the input space to a higher-dimensional feature space so that the sample is linearly separable in the new space. The kernel function $k(x, x_i)$ and Lagrange equation are introduced to solve the high-dimensional space classification hyperplane.

$$f(x) = \sum_{i=1}^{N} \alpha_i y_i k(x, x_i) + b_s \tag{3}$$

where α_i is the Lagrange multiplier.

In the SVM algorithm, different kernel functions can be used for sample mapping, which is a key factor in determining the performance of the SVM. Common kernel functions include linear, polynomial, and radial basis functions. In this paper, the radial basis function[30] is selected as the kernel function for the algorithm, and its form is as follows.

$$k(x, x_i) = \exp\left(-\gamma ||x - x_i||^2\right) \tag{4}$$

where γ is the influence factor of the training sample.

An overview of PSO algorithm

The PSO algorithm is an evolutionary computation technique inspired by the foraging behavior of bird swarms. In this algorithm, a swarm of particles explores the search space to find the optimal solution, mimicking the way birds locate food. Each particle does not know the exact location of the optimal solution but can assess its proximity based on its current position. The best strategy is to focus the search around the most promising particles, gradually guiding the swarm toward the optimal solution.

In PSO, each particle represents a potential solution with a fitness value determined by the objective function. The movement of particles is governed by their velocity, which is dynamically adjusted based on their own best-known position and the best-known position of the swarm. By iteratively updating their positions and velocities, particles collectively explore the solution space and converge toward the optimal solution.

In traditional PSO algorithms, the process is primarily divided into two key components: initializing the particle population and iteratively updating their positions and velocities. 5The algorithm begins by randomly generating an initial position space X and assigning a random velocity V to each particle. The movement vector of each particle plays a crucial role in updating the population, as illustrated in Eq. (). This initialization phase lays the foundation for the subsequent iterative search and optimization process.

$$\begin{gathered} X = rand(N, dim) * (ub - lb) + lb \\ V = zeros(N, \dim) \end{gathered} \tag{5}$$

where N denotes the population size, *dim* represents the dimensionality of the population, *ub* signifies the upper boundary of the search space, and *lb* indicates the lower boundary. The algorithm subsequently locates the optimal solution via iterative refinement. During each iteration, particles update their states by monitoring two extrema. The first extremum corresponds to the best solution discovered by each particle, termed the personal best or individual extremum. Each particle's optimal position is denoted as *pBest*, as exemplified in Eq. (6).

$$pBest_i = (pBest_{i,1}, pBest_{i,2}, pBest_{i,3} \cdots pBest_{i,dim}) \tag{6}$$

In this context, the variable $pBest_i$ represents the optimal position identified by the i-th particle within the current population. The procedure for updating this position is elaborated in Eq. (7).

$$pBest_i(t+1) = \begin{cases} X_i(t+1), & if\ f(X_i(t+1)) < f(pBest_i(t)) \\ pBest_i(t), & otherwise \end{cases} \tag{7}$$

Within the equation, $X_i(t+1)$ denotes the updated position of the individual post-update, $pBest_i\,(t+1)$ signifies the most favorable position found by the particle following the $t+1$ update, and $f(.)$ denotes the fitness evaluation function employed to compute the fitness of each individual. The alternate extremum, designated as the global optimum *gBest*, represents the optimal solution found by the whole population. Regarding the entire particle ensemble, each particle's current search position constitutes a potential candidate for the global optimum, as illustrated in Eq. (8).

$$gBest = (gBest_1, gBest_2, gBest_3 \cdots gBest_{dim}) \tag{8}$$

The methodology for updating *gBest* is delineated in Eq. (9).

$$gBest = \begin{cases} X_i\,(t+1), & if\ f\,(X_i\,(t+1)) < f\,(gBest) \\ gBest, & otherwise \end{cases} \tag{9}$$

where *gBest* signifies the global optimum. In the PSO algorithm, both individual and population updates rely on the velocity vector of each particle, which is detailed in Eq. (10).

$$V_i = (V_{i,1}, V_{i,2}, V_{i,3} \cdots V_{i,dim}) \tag{10}$$

Leveraging this vector, the PSO algorithm determines each particle's update direction and velocity, ensuring efficient exploration of the search space for the optimal solution. The updated methodology is outlined in Eq. (11).

$$V_{i,j}\,(t+1) = w \cdot V_{i,j}\,(t) + c_1 \cdot rand \cdot (pBest_{i,j}\,(t) - X_{i,j}\,(t)) + c_2 \cdot rand \cdot (gBest_j - X_{i,j}\,(t)) \tag{11}$$

where c_1 and c_2 denote the particle learning factors, and w signifies the inertia weight. To enable particles to operate within bounded limits and enhance the efficacy of optimal position discovery, the PSO algorithm imposes velocity constraints, as delineated in Eq. (12).

$$V_{i,j} = \begin{cases} V_{max} V_{i,j} > V_{max} \\ -V_{max} V_{i,j} < -V_{max} \end{cases} \tag{12}$$

Ultimately, the particle position update method is specified in Eq. (13). Through this, the particle swarm iterates persistently until achieving the requisite precision or attaining the maximum iteration count.

$$X_{i,j}\,(t+1) = X_{i,j}\,(t) + V_{i,j}\,(t+1) \tag{13}$$

The proposed OLDPSO algorithm

Opposition-based Latin squares sampling initialization

The initialization of the PSO algorithm plays a critical role, significantly influencing both the convergence speed and trajectory throughout the optimization process. Initially, an opposition-based learning[31] population strategy is employed. Opposition-based learning, an advanced optimization technique, generates counterparts to the original particles to expand the search domain and improve the quality of solutions. By applying opposition-based learning, each particle produces its antithetical counterpart based on predefined rules or algorithms, contrasting with its original characteristics, thus enabling a more comprehensive exploration of the search space. This approach ensures that the algorithm captures a broader diversity of potential solutions, thereby increasing the probability of identifying the global optimum.

Most current research employing an opposition-based learning strategy generates the entire opposition-based learning population. In contrast, this paper introduces a novel approach by generating only half of the opposition-based learning population, which offers three distinct advantages over generating the complete population: First, this approach enhances computational efficiency by avoiding the surplus of redundant searches associated with generating opposite individuals, thereby minimizing superfluous computations through halving the population. Second, it optimizes resource utilization by maximizing the efficiency of available resources when generating only half of the opposition-based learning population. Third, generating the entire set of opposite individuals could overly complicate the population, potentially impairing search performance, whereas generating just half avoids this unnecessary complexity. By adopting this half-generation strategy, the algorithm navigates the search space with greater efficiency, thereby enhancing both search effectiveness and optimization precision.

According to Eq. (14), half of the opposition-based learning population is generated, with *midnum* defined in Eq. (15). Here, X_{rev} denotes the opposition-based learning population, *dim* represents the population dimension, *lb* is the lower bound of the search space, *ub* is the upper bound, *X* refers to the original population, and *SearchAgents* signifies the number of original populations.

$$X_{rev} = rand\,(midnum, dim)\,.*\,(lb - ub) - X\,(1 : midnum, :) \tag{14}$$

$$midnum = floor(SearchAgents/2) \tag{15}$$

In the opposition-based learning strategy, the singular transformation of opposite points can result in some individuals persistently targeting the local optimum during later iterations. Within the PSO algorithm, a particle's descent into a local optimum tends to attract others nearby, culminating in an effect that is pronounced at the

outset but diminishes over subsequent iterations. Consequently, relying solely on opposition-based learning for population initialization does not yield optimal results.

To mitigate this limitation, this paper incorporates a Latin square sampling mechanism[32] into its opposition-based learning framework. Latin square sampling, a stratified random technique, cleverly segments data based on multiple parameter distributions, enabling random sampling within each segment to approximate universal randomness. A primary benefit of Latin square sampling is its ability to garner greater insights with reduced repetition. By employing judicious stratification and random sampling tactics, the approach ensures sample representativeness while minimizing superfluous repetition, thus enhancing sampling efficiency. Moreover, Latin square sampling contributes to a more diffuse and evenly spread distribution of data. This paper employs a modified Latin square sampling mechanism to initialize search agent positions within a multidimensional search space. Traditional Latin square sampling ensures a uniform distribution of samples along each dimension, with uniqueness maintained for each sample within that dimension. However, this study introduces randomness via the addition of *rand*( ), ensuring positions are not rigidly grid-aligned. The algorithm initially creates a uniformly randomized index, as depicted in Eq. (16). Subsequently, the step size is calculated, as illustrated in Eq. (17). Finally, Latin square sampling values for the current dimension are populated, as shown in Eq. (18). Here, *idx* denotes the index sequence, *step* the step size, *i* the current dimension, and *j* the current particle.

$$idx = randperm\left(Search\ Agents\right) \tag{16}$$

$$step = \frac{ub\left(i\right) - lb\left(i\right)}{Search\ Agents} \tag{17}$$

$$X_{rev}\left(j,i\right) = lb\left(i\right) + step * \left(idx\left(j\right) - rand\left(\ \right)\right) \tag{18}$$

In summary, the specific steps for population initialization using the opposition-based Latin squares sampling initialization method are as follows: first, randomly generate a population of size *N*. Next, create an opposition-based learning population of size *N*/2 and apply Latin square sampling to it. Merge these two populations to form a mixed population of size 3*N*/2. Calculate the fitness function values for each individual, and select the top *N* individuals based on their fitness values to form the final population. The population initialization's procedural flow is depicted in Fig. 2. Integrating an opposition-based learning mechanism with Latin square sampling to create this opposition-based Latin square sampling initialization scheme notably decreases the likelihood of premature convergence in the PSO algorithm, thereby preventing early entrapment in local optima.

Dynamic inertia weights and learning factors

Conventional PSO algorithms typically apply uniform inertia weights and learning factors across all particles, overlooking the need for individualized tuning. This oversight can significantly hinder the maintenance of population diversity. To address this limitation, this paper introduces a strategy of dynamic inertia weights and learning factors. This approach dynamically adjusts the control parameters of the particle swarm from both vertical and horizontal dimensions. By doing so, the adaptive mechanism enables particles to focus on tasks that are optimally suited to their current stage, effectively mitigating conflicts between global exploration and local exploitation during parameter setting.

The dynamic inertia weights strategy employs a cosine decay strategy for adjusting the inertia weights. A hallmark of this approach is its maintenance of a high inertia weight during the initial phase of the algorithm, facilitating a swifter exploration of the search space. However, as the algorithm approaches the optimum, the inertia weight decreases, allowing for finer adjustments to the search direction, thereby improving the convergence rate and precision. This methodology strikes an optimal equilibrium between global and local

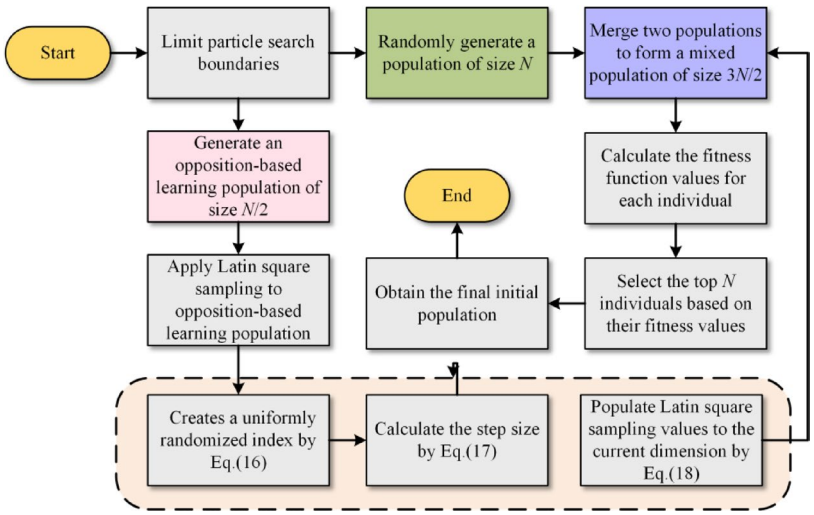


Fig. 2. The special framework of population initialization.

Input: Population size (N), maximum evaluation number ($MaxFEs$), lower bound of search space(lb), upper bound of search space(ub), dimension (dim), and the fitness function $fobj(x)$

Output: the best location ($bestPos$)

Initialize the velocity vector: $V = zeros(N, dim)$

Initialize the optimal position: $pBest = zeros(N, dim)$

Initialize grade of the current individual: $pBestScore = zeros(N)$

Initialize position vector: $gBest = zeros(1, dim)$

Initialize score for the best location: $gBest_score = inf$

Initialize population:

Check if ub and lb are scalars

Initialize a random population X and particle velocity V by Eq. (5)

Generate half of the opposition-based learning population X_{rev}

For i = 1: dim

Generate a randomly arranged index with uniform distribution idx

Calculate $step$

Fill the Latin square sampling values for the current dimension

End

Merge populations into X_total

Calculate the objective function value and ascending the objective function value

Obtain the final population with the best fitness function values for the top N fitness functions

Initialize the parameters: t, FEs

While $(FEs < MaxFEs)$

For i = 1: size(X,1)

Return the particles that go beyond the boundaries of the search space

Calculate the objective function for each search particle

$$FEs = FEs + 1$$

Update the locations and scores of $pBest_i$ and $gBest$

End for

Finding the maximum and minimum fitness values

Calculate the average fitness value $\bar{f}$

Initialize the parameters: $\alpha, \beta, w_{max}, w_{min}$

Update the value of w_t by Eq. (19)

Update the value of w_f by Eq. (20)

Update the adaptive weight w by Eq. (21)

Update the learning factors c_1, c_2 by Eq. (22)

For i = 1: size(X,1)

For j=1: size(X,2)

Update the velocity vector $V_{i,j}$ by Eq. (11)

Update the location of particles by Eq. (13)

End for

End for

$t = t + 1$

End while

Return $bestPos$

Algorithm 1. Pseudocode for the OLDPS

searches, thereby boosting the algorithm's overall efficacy. The inertia parameter w_t, influenced by the number of iterations, is formulated as a function of *t*, depicted in Eq. (19), where T represents the total number of iterations. The parameters w_{max} and w_{min} were initialized with fixed values of 0.9 and 0.4, respectively.

$$w_t = (w_{max} - w_{min}) \cdot cos\frac{\pi\, t}{2T} + w_{min} \tag{19}$$

Following each iteration, compute the fitness values of the particles. Upon discarding particles with infinite fitness values, identify the maximum f_{max} and minimum f_{min} fitness values among the survivors, and determine the mean fitness value $\overline{f}$. The inertia parameter w_f , influenced by the fitness value f_i of each particle, is defined as a function of *f*, as delineated in Eq. (20).

$$w_f = (w_{max} - w_{min}) \cdot \left(\frac{f_i - \overline{f}}{f_{max} - f_{min}} + 1\right) + w_{min} \tag{20}$$

Considering the impact of iteration count *t* and fitness value *f* on the inertia weight, *w* is presented in Eq. (21). Here, α and β serve as proportionality constants to modulate the effects of *t* and *f* on *w*.

$$w = \alpha \cdot w_t + \beta \cdot w_f \tag{21}$$

Typically, the PSO algorithm includes two learning factors: c_1 for self-awareness and c_2 for social cognition. These are crucial parameters within the algorithm. The choice of learning factors significantly influences PSO algorithm performance. Setting c_1 and c_2 too high can result in premature convergence or unstable searches; conversely, setting them too low can lead to slow convergence or becoming trapped in local optima. Ultimately, the learning factors c_1 and c_2 are derived from the inertia weight, as illustrated in Eq. (22).

$$\begin{cases} c_1 = \quad 3 \cdot w - 0.7 \\ c_2 = -3 \cdot w + 3.2 \end{cases} \tag{22}$$

OLDPSO algorithm

To comprehensively enhance the performance of PSO, this paper innovatively integrates the above two optimization strategies, and finally proposes the OLDPSO algorithm. Initially, we introduce an opposition-based Latin square sampling mechanism to refine PSO's initialization. This enhances both the initial population's quality and its spatial traversal efficiency. Subsequently, dynamic inertia weights and learning factors are introduced to maintain diversity while boosting global and local search efficacy simultaneously. Grounded in these strategies, the proposed OLDPSO algorithm is outlined in Algorithm 1, with its procedural flow depicted in Fig. 3. The proposed algorithm optimizes both initialization and search phases, significantly enhancing the overall performance of PSO.

Computational complexity of OLDPSO

The time complexity of the OLDPSO algorithm involves two main phases: initialization and the main optimization loop.

During initialization, generating *N* particles each with *dim* dimensions results in $O(N \times dim)$. Additionally, creating about $N/2$ opposition-based learning particles using random permutation and Latin square sampling methods also contributes $O(N \times dim)$. Evaluating and sorting all particles to select the top *N* optimal ones incurs $O(N \times \log N)$. Thus, the total initialization complexity is $O(N \times dim + N \times \log N)$.

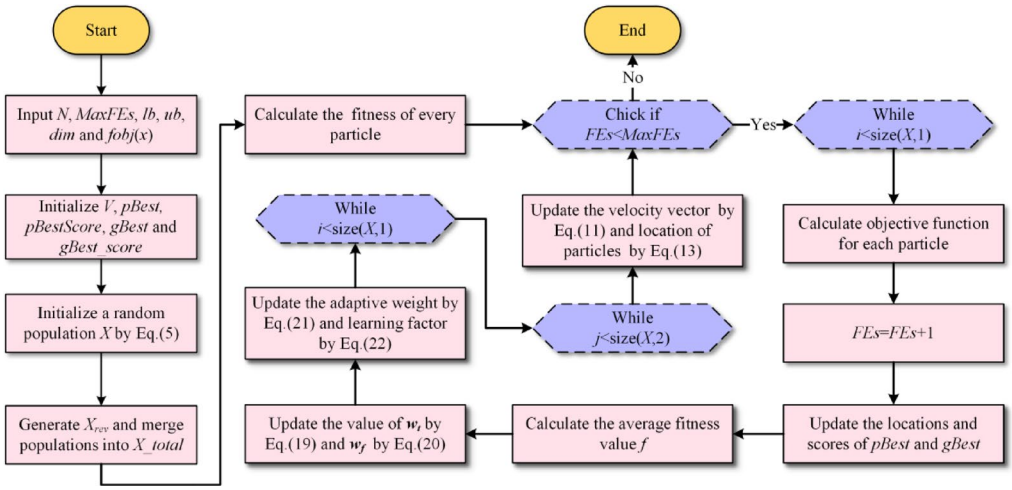


Fig. 3. The special framework of the proposed OLDPSO.

In the main optimization loop, the termination condition is based on the number of evaluations (*FEs*) reaching the maximum allowed (*MaxFEs*). Each iteration includes boundary checking for *N* particles across *dim* dimensions $O(N \times dim)$, evaluating and updating personal bests (*pBest*) and global best (*gBest*) $O(N)$, calculating fitness statistics $O(N)$, and updating velocities and positions $O(N \times dim)$. Given that each iteration processes *N* evaluations, the total number of iterations is approximately *MaxFEs*/*N*, leading to an overall time complexity for the main optimization loop of $O(MaxFEs \times dim)$.

Combining both phases, the overall time complexity of the OLDPSO algorithm is $O(N \times dim + N \times logN + MaxFEs \times dim)$. In most practical scenarios, *MaxFEs* significantly exceeds $N \times logN$ and $N \times dim$, making the dominant term $O(MaxFEs \times dim)$. This indicates that the algorithm's runtime scales linearly with both *MaxFEs* and the dimensionality of the problem (*dim*), making it well-suited for high-dimensional optimization tasks requiring extensive evaluations.

Data and proposed OLDPSO-SVM model

Data collection

The AD Neuroimaging Initiative (ADNI), launched in 2003, is a collaborative project that includes the National Institute on Aging (NIA), the National Institute of Biomedical Imaging and Bioengineering (NIBIB), and the U.S. Food and Drug Administration (FDA), among other institutions[33]. Its key goal is to facilitate early diagnosis and encourage the development of innovative treatments for AD. As a result, ADNI has become a critical resource for researchers by offering vast datasets and promoting collaborative efforts in research. Further information about this initiative can be accessed at http://adni.loni.usc.edu/ADNI. In this study, we analyzed 414 participants based on the baseline MRI scans provided by ADNI. These subjects were divided into four distinct categories: 103 individuals diagnosed with AD, 95 with MCI who progressed to AD (MCI-C) within 18 months, 104 with MCI who did not progress to AD (MCI-NC) during the same period, and 112 NC subjects. A comprehensive summary of demographic information for all participants is presented in Table 1.

Image preprocessing and feature extraction

To facilitate feature extraction, a standardized image preprocessing approach is implemented, tailored to the characteristics of the acquired MRI images. The workflow for image preprocessing and feature extraction is illustrated in Fig. 4.

Figure 4 outlines the data preprocessing workflow based on the ADNI-pipeline method detailed in[34] comprising three main steps: (1) post-acquisition correction; (2) B1-intensity variation non-uniformity correction; and (3) intensity non-uniformity correction for 3T MRI scans. This study utilized the CAT12 toolbox (http://www.neuro.uni-jena.de/cat/) within the SPM12 framework (https://www.fil.ion.ucl.ac.uk/spm/software/spm12/) for VBM analysis, which is known for its superior sensitivity in detecting GM changes compared to VBM8 with SPM8[35]. The CAT12-VBM preprocessing pipeline encompasses normalization, skull stripping, segmentation, and smoothing, with each MRI classified into GM, WM, and cerebrospinal fluid.

Upon completion of the preprocessing steps, voxel features representing GM and WM volumes were extracted using the AAL template, which is an automatic anatomical atlas described in[36]. Each subject's brain was divided into 90 regions of interest (ROIs), as outlined in Table 2, where R1-R90 indicates the ROIs, with odd numbers for the left hemisphere and even numbers for the right. For each ROI, both GM and WM volumes were calculated as separate features. Thus, combining GM and WM volumes for all ROIs, there are a total of 180 features. These volumes were then used as features for experiments.

The proposed OLDPSO-SVM model description

As illustrated in Fig. 5, the process begins with the extraction of GM and WM volumetric features, which are subsequently scaled to the range of [-1,1] and used as input data for the SVM. The feature subset population is then initialized, with particle dimensions corresponding to the number of features. For each particle, features are selected based on their binary encoding. The SVM evaluates the selected feature subsets using 10-fold cross-validation, and the fitness of each particle is determined by the classification performance of the SVM. The proposed OLDPSO algorithm optimizes feature selection by iteratively updating the particles until the convergence criteria are satisfied. Ultimately, the best feature subset is identified, and its classification performance is evaluated on the testing dataset.

In the OLDPSO-SVM model, binary encoding is employed to represent the solution space of the wrapper-based feature selection problem. Specifically, each dimension of a particle corresponds to a feature, where a binary value indicates whether a feature is selected or not. A value of "1" signifies that the feature is included in the subset, while "0" indicates that the feature is excluded. To convert the continuous position of a particle into a discrete binary representation, various TFs are introduced, including both S-shaped and V-shaped functions.

Diagnosis	Number	Age	Gender (M/F)	MMSE	Education	CDR
AD	103	75.32 ± 7.59	52/51	23.23 ± 1.97	14.6 ± 3.2	0.75 ± 0.25
NC	112	76.44 ± 5.35	55/57	29.27 ± 0.89	16.25 ± 2.78	0 ± 0
MCI-C	95	74.84 ± 7.08	56/39	26.49 ± 1.79	15.65 ± 2.92	0.5 ± 0
MCI-NC	104	74.66 ± 7.78	52/52	27.21 ± 1.65	15.8 ± 3.11	0.5 ± 0

Table 1. The comprehensive demographic overview of study participants. MMSE: the Mini-Mental State Examination. CDR: the Clinical Dementia Rating. The data are presented as mean values ± standard deviation.

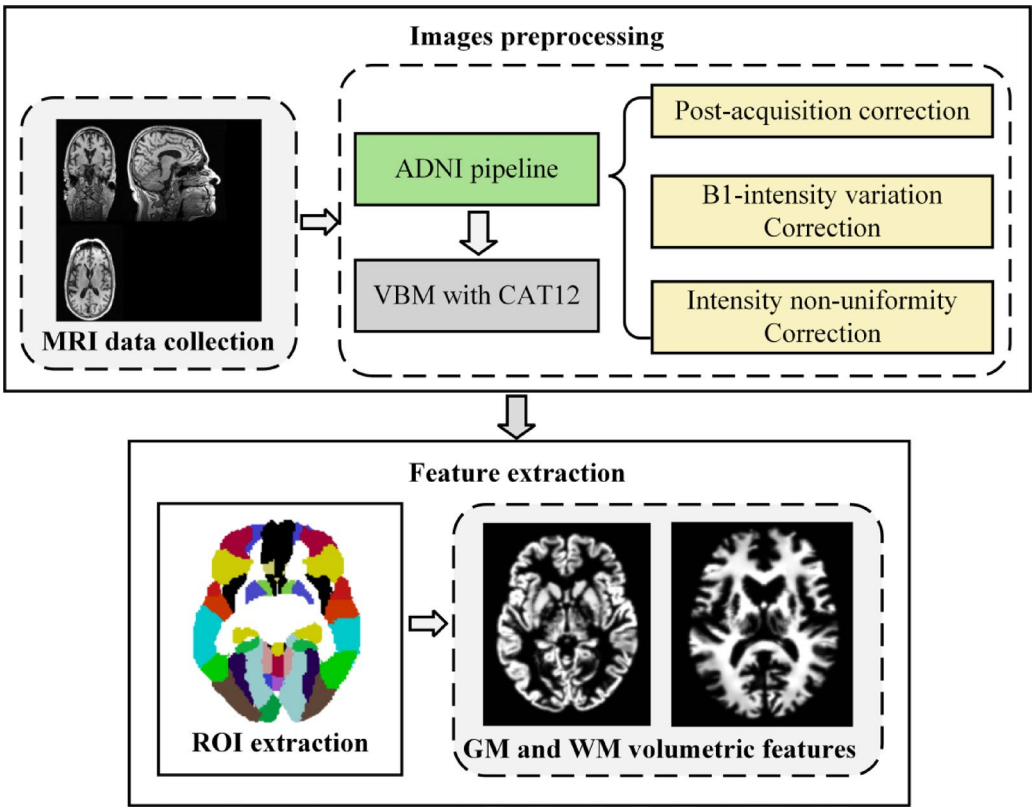


Fig. 4. Flowchart of MRI data preprocessing and feature extraction. MRI data were obtained from the ADNI database (http://adni.loni.usc.edu/).

Different TFs are applied to address various problems, as each TF offers distinct mapping characteristics that are suited to different optimization landscapes. This allows the model to adapt to the nature of the feature selection task and improve the search process by selecting the most appropriate transformation. These TFs play a key role in transforming continuous values into binary ones, ensuring efficient feature selection. The specific formulas for these TFs are shown in Table 3.

$$Fitness = w_1 \cdot ErrorRate + w_2 * \frac{number\ of\ selected\ features}{total\ number\ of\ features} \tag{23}$$

where $w_1 = 0.95$ and $w_2 = 0.05$, prioritizing classification accuracy (95%) while encouraging dimensionality reduction (5%) without significantly compromising performance[37].

Several key evaluation metrics are employed to assess the performance of feature selection and classification. Accuracy measures the overall proportion of correctly classified instances, while Sensitivity evaluates the model's ability to correctly identify positive instances. Specificity assesses the ability to correctly identify negative instances, and Precision focuses on the proportion of predicted positives that are actually correct. The Matthews Correlation Coefficient (MCC) offers a balanced assessment by considering all true and false classifications, making it particularly useful for imbalanced datasets. Lastly, the F-measure balances Precision and Sensitivity by calculating their harmonic mean, providing a single, robust metric for performance evaluation. The mathematical definitions of these evaluation metrics are as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{24}$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} \tag{25}$$

$$Specificity = \frac{TN}{TN + FP} \tag{26}$$

$$Precision = \frac{TP}{TP + FP} \tag{27}$$

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP) \times (TP + FN) \times (TN + FP) \times (TN + FN)}} \tag{28}$$

90 ROIs	Brain region
R1, R2	Precentral gyrus
R3, R4	Superior frontal gyrus (dorsal)
R5, R6	Orbitofrontal cortex (superior)
R7, R8	Middle frontal gyrus
R9, R10	Orbitofrontal cortex (middle)
R11, R12	Inferior frontal gyrus (opercular)
R13, R14	Inferior frontal gyrus (triangular)
R15, R16	Orbitofrontal cortex (inferior)
R17, R18	Rolandic operculum
R19, R20	Supplementary motor area
R21, R22	Olfactory
R23, R24	Superior frontal gyrus (medial)
R25, R26	Orbitofrontal cortex (medial)
R27, R28	Rectus gyrus
R29, R30	Insula
R31, R32	Anterior cingulate gyrus
R33, R34	Middle cingulate gyrus
R35, R36	Posterior cingulate gyrus
R37, R38	Hippocampus
R39, R40	Parahippocampal gyrus
R41, R42	Amygdala
R43, R44	Calcarine
R45, R46	Cuneus
R47, R48	Lingual gyrus
R49, R50	Superior occipital gyrus
R51, R52	Middle occipital gyrus
R53, R54	Inferior occipital gyrus
R55, R56	Fusiform gyrus
R57, R58	Postcentral gyrus
R59, R60	Superior parietal gyrus
R61, R62	Inferior parietal lobule
R63, R64	Supramarginal gyrus
R65, R66	Angular gyrus
R67, R68	Precuneus
R69, R70	Paracentral lobule
R71, R72	Caudate
R73, R74	Putamen
R75, R76	Pallidum
R77, R78	Thalamus
R79, R80	Heschl gyrus
R81, R82	Superior temporal gyrus
R83, R84	Temporal pole (superior)
R85, R86	Middle temporal gyrus
R87, R88	Temporal pole (middle)
R89, R90	Inferior temporal

Table 2. 90 ROIs and their corresponding brain regions.

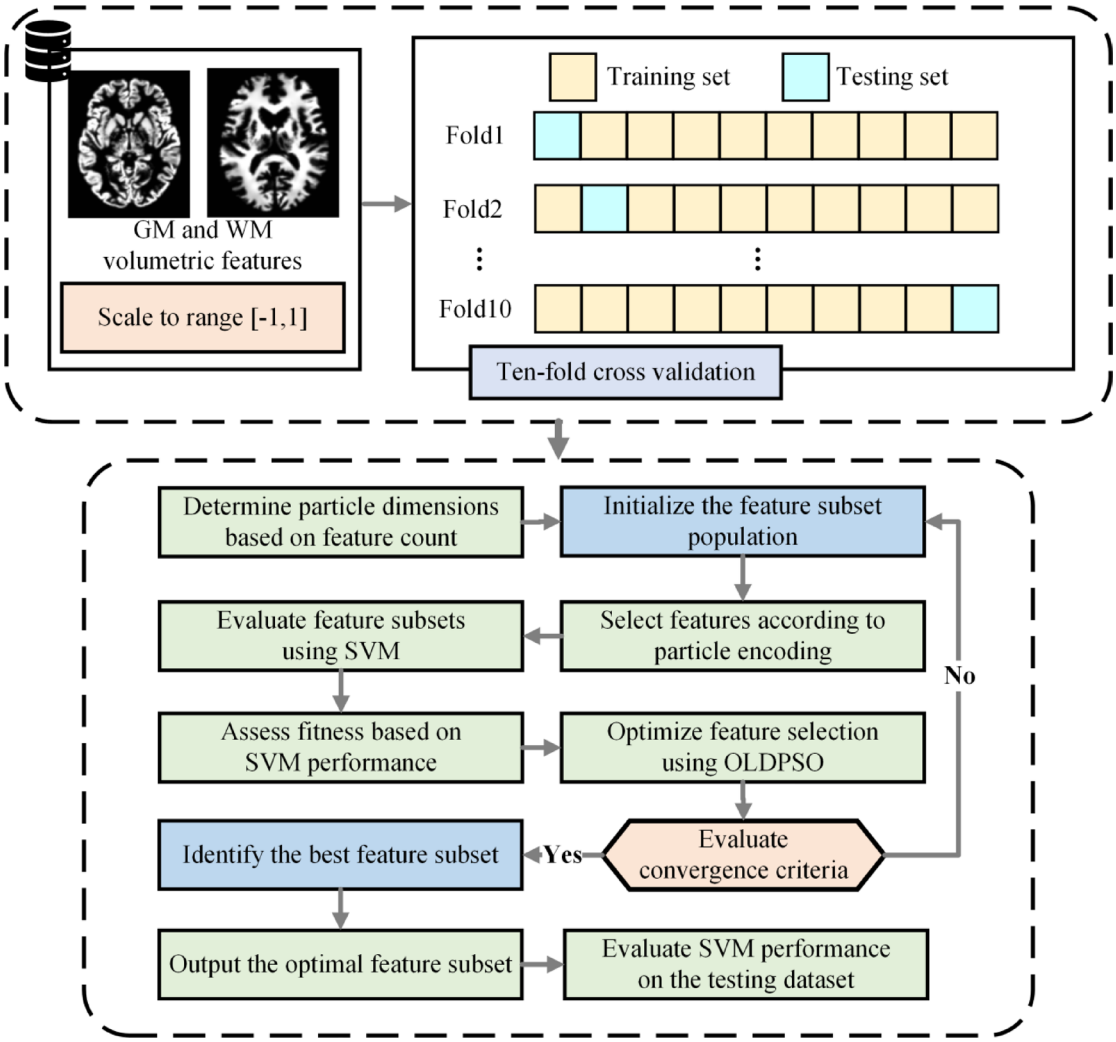


Fig. 5. Feature selection process using OLDPSO-SVM was created based on MRI data from ADNI database (http://adni.loni.usc.edu/).

S-shaped TF		V-shaped TF	
Style	Formula	Style	Formula
TFs			
S1	$T(x) = \frac{1}{1+e^{-2x}}$	V1	$T(x) = \left\vert \frac{2}{x}\int_0^{(\sqrt{\pi}/2)X} e^{-t^2}dt \right\vert$
S2	$T(x) = \frac{1}{1+e^{-x}}$	V2	$T(x) = \vert \tanh x \vert$
S3	$T(x) = \frac{1}{1+e^{-x/2}}$	V3	$T(x) = \left\vert x/\sqrt{1+x^2} \right\vert$
S4	$T(x) = \frac{1}{1+e^{-x/3}}$	V4	$T(x) = \left\vert \frac{2}{x}\arctan\left(\frac{\pi}{2}x\right) \right\vert$

Table 3. S-shaped and V-shaped TFs for feature selection. In the OLDPSO-SVM model, the fitness function considers both the classification error rate and the number of selected features, balancing the goal of dimensionality reduction with maintaining classification accuracy. The fitness value consists of the SVM error rate and the ratio of selected features, as follows:

PSO variants	Parameter configurations	Reference
OLDPSO	$V_{max} = 6; w_{max} = 0.9; w_{min} = 0.4; c_1 = 3 * w - 0.7; c_2 = -3 * w + 3.2$	–
ALCPSO	$w = 0.4;\ c1 = c2 = 2; lifespan = 60;\ T = 2;\ pro = 1/n$	[38]
CGPSO	$c1 = c2 = 2; W_max = 0.9; W_min = 0.2; V_max = 6$	[39]
FST_PSO	$V_{max} = 6; V_{min} = -6$	[40]
ASCA_PSO	$V_{max} = 6; w_{max} = 0.9; w_{min} = 0.2$	[41]
HCLPSO	interval = ub − lb; $V_{max} = 0.2$*interval; $V_{min} = -V_{max}$	[42]
SCA_PSO	$V_{max} = 6; V_{min} = -6$	[43]

Table 4. The parameter configurations employed for comparing PSO variants.

Parameter	description	Value
N	Population size	30
Dim	Problem dimension	30
$MaxFEs$	Maximum evaluation times	$Dim \times 10^4$
$Flods$	Number of independent runs	30

Table 5. Common parameter settings.

$$\text{F - measure} = 2 \times \frac{Precision \times \text{Sensitivity}}{Precision + \text{Sensitivity}} \tag{29}$$

where *TP* represents true positives, *TN* represents true negatives, *FP* represents false positives, and *FN* represents false negatives.

Experimental results of the proposed OLDPSO

This experiment consists of two main components aimed at comprehensively evaluating the optimization performance of OLDPSO. First, we conducted an ablation study, where OL represents the initialization scheme based on opposition-based Latin square sampling, and D denotes the dynamic inertia weights and learning factors. In this experiment, we systematically assessed the practical impacts of the OL and D strategies. Subsequently, we performed a visual analysis of the optimization trajectories of OLDPSO throughout the process. Furthermore, to emphasize the superiority of OLDPSO, we compared its performance with several state-of-the-art improved PSO algorithms, including PSO with an aging leader and challengers (ALCPSO)[38], PSO with chaotic and gaussian local search (CGPSO)[39], fuzzy self-tuning PSO (FST_PSO)[40], adaptive sine cosine optimization algorithm integrated with PSO (ASCA_PSO)[41], heterogeneous comprehensive learning PSO (HCLPSO)[42], and sine-cosine PSO (SCA_PSO)[43]. Through this comparative analysis, we reveal the significant advantages and superior performance of the OLDPSO algorithm relative to these complex methodologies.

Table 4 enumerates the parameter configurations employed for comparing PSO variants. Experiments were executed using the CEC 2017[44] benchmark suite, encompassing four categories: unimodal, multimodal, hybrid, and composite functions. The CEC 2017 benchmark was selected due to its comprehensive and challenging set of test functions, which facilitate a rigorous evaluation of optimization algorithms across diverse problem landscapes, thereby ensuring a robust assessment of the performance of the PSO variants. Table 5 offers detailed specifications of the shared parameters in the experimental setup. To uphold comparability impartiality, each algorithm underwent independent execution 30 times. Moreover, this paper furnishes a meticulous analysis of experimental outcomes across five pivotal metrics: average value result (AvgR), standard deviation result (StdR), Wilcoxon Signed-Rank Test (WSRT)[45], Friedman test (FRT)[46], and convergence rate. These analyses afford a thorough and nuanced insight into the outcomes of our experiments.

Ablation study

This section presents empirical analysis to validate the benefits of incorporating OL and D strategies in enhancing PSO algorithm performance. We implemented four distinct PSO variants, as detailed in Table 6, where OL represents opposition-based Latin square sampling initialization and D denotes dynamic inertia weights and learning factors. In the table, a "1" indicates the inclusion of a mechanism, while a "0" indicates its absence. This design clearly illustrates the interplay among the various mechanisms, providing insights into their impact on algorithmic performance.

To evaluate the efficacy of the four PSO variants, this study employed the CEC 2017 benchmark suite. The FRT was conducted to determine the mean rank of each variant, enabling a comparison of the effects of different mechanism combinations. Table 7 presents the final ranks and average ranking values (ARV) for the four algorithms examined. The ARV reflects the mean rank obtained from FRT across 30 CEC 2017 benchmarks. In the table, the "R" symbols indicate statistically significant performance differences compared to OLDPSO under WSRT, with "+", "-", and "=" denoting that OLDPSO's performance is significantly superior, inferior, or

	OL	D
OLDPSO	1	1
OLPSO	1	0
DPSO	0	1
PSO	0	0

Table 6. Comparison of PSO variants utilizing OL and D strategies.

Algorithms	Ranking	ARV	*R+/R-/R=*
OLDPSO	1	1.677222	~
OLPSO	3	3.157778	28/0/2
DPSO	2	1.822778	18/7/5
PSO	4	3.342222	30/0/0

Table 7. Rankings and performance metrics for OLDPSO and other PSO variants.

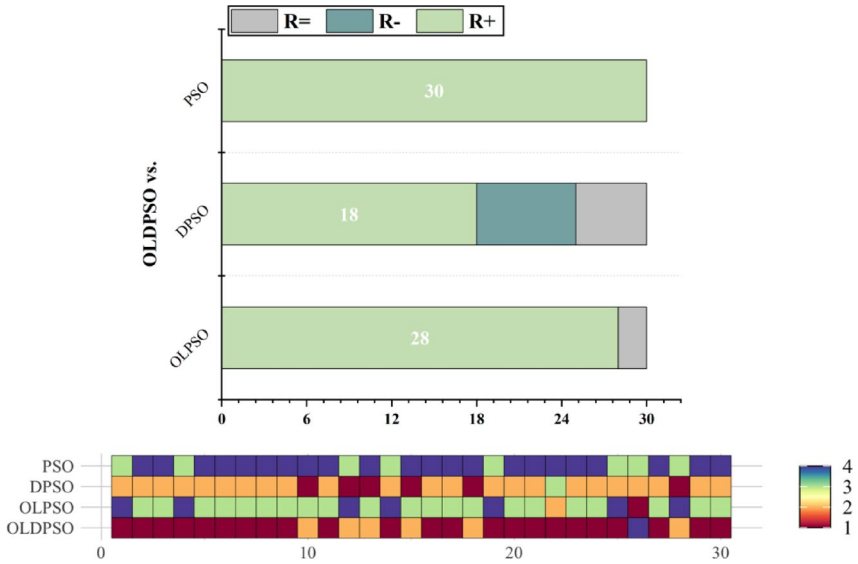


Fig. 6. Performance comparison of OLDPSO and alternative PSO algorithms.

equivalent to that of the other algorithms, respectively. Notably, OLDPSO achieved the highest ARV of 1.677222, followed by DPSO and OLPSO with ARVs of 1.822778 and 3.157778, respectively. The standard PSO recorded an ARV of 3.342222, indicating that OLDPSO outperformed the other variants in solving the majority of test functions, thereby validating the effectiveness of the optimization strategies employed.

Moreover, Fig. 6 offers a comprehensive visual comparison of OLDPSO's performance against three alternative algorithms. The top section presents the "R+/R-/R=" results, illustrating that OLDPSO consistently outperforms the other algorithms across most test functions. In particular, OLDPSO surpasses the baseline PSO algorithm in all cases, demonstrating its superior optimization capabilities. Additionally, the heatmap based on 30 runs shows rankings for each algorithm, where dark red signifies the top rank (1st place) and dark blue indicates the lowest rank (4th place). The heatmap reveals that OLDPSO frequently ranks first, while the standard PSO often ranks fourth, further underscoring OLDPSO's strong performance compared to the other variants.

Figure 7 presents the convergence trajectories of four algorithms, namely OLDPSO, OLPSO, DPSO, and PSO, on nine representative benchmark functions, highlighting OLDPSO's superior speed and efficiency in reaching optimal solutions. The horizontal axis shows the *FEs*, up to a limit of 300,000. The vertical axis indicates the best fitness score obtained so far during the search process. Among the four algorithms, OLDPSO consistently demonstrates superior performance across all benchmark functions. Notably, the enhanced convergence speed of OLDPSO is evident across a range of benchmark functions, including both simpler and more complex multimodal functions such as F5, F6, F8, F9, F16, F20, F21, F22, and F23. When compared to the other PSO variants, OLDPSO exhibits the fastest convergence rate and achieves the most competitive fitness values. This performance improvement is largely attributed to the integration of an opposition-based Latin square sampling initialization mechanism, which significantly reduces the risk of premature convergence, helping to prevent the

algorithm from becoming trapped in local optima too early in the search process. Moreover, the use of dynamic inertia weights and adaptive learning factors strategies further enhances the algorithm's ability to balance global exploration and local exploitation, leading to more efficient searches and better overall results.

To further evaluate the performance of the algorithms, Fig. 8 presents the box plots of the results for functions F1, F3, F8, F9, F14, and F30. The aim is to analyze the variability and stability of the optimization results across these benchmark functions. For F1, F3, and F30, both OLDPSO and DPSO show similar strong performance, as indicated by their smaller box sizes, suggesting lower variability and better stability in their results. In these cases, the narrow box sizes, which sometimes appear almost as straight lines, demonstrate the precision of the optimization process, underscoring the effectiveness of the dynamic inertia weights mechanism. However, OLDPSO outperforms the other algorithms in F8, F9, and F14, with DPSO performing slightly behind. Overall, OLDPSO consistently exhibits superior performance, with its box plots displaying narrower ranges across all six functions, highlighting its strong optimization capability and stability compared to the other PSO variants.

Visual and qualitative analysis of OLDPSO

This section presents a qualitative analysis of the proposed OLDPSO algorithm to demonstrate its performance more comprehensively. The population size is set to 20, the maximum number of iterations to 1000, and the dimensionality to 30. Figure 9 illustrates the qualitative results of OLDPSO applied to unimodal, multimodal, hybrid, and composite functions.

The first column (a) displays the 3D plots of selected benchmark functions from the 30 IEEE CEC 2017 functions, which visually illustrate the shape and complexity of the functions, helping to understand the algorithm's performance on different functions. The second column (b) presents the search trajectories of standard PSO, while the third column (c) shows those of OLDPSO, with red dots marking the optimal solutions and black dots representing individual trajectories. Comparing (b) and (c), PSO individuals are more scattered, while OLDPSO individuals converge tightly around the optimal solutions, showing improved optimization. The fourth column (d) illustrates the trajectory of OLDPSO in the first dimension throughout the iterations, showing the fluctuation of individuals' positions with respect to time. Initially, the oscillation amplitude is large, but it gradually decreases and stabilizes as the search progresses, indicating convergence toward the optimal solution.

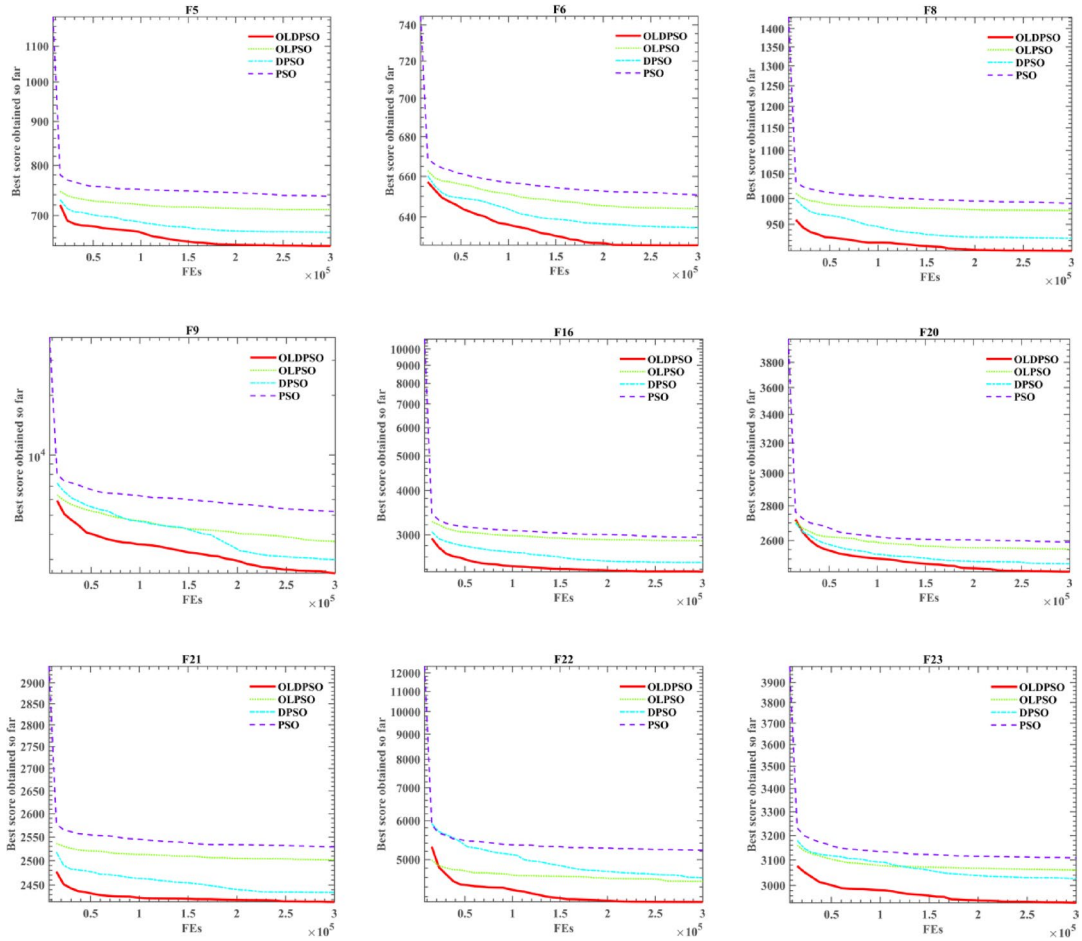


Fig. 7. Convergence performance comparison of OLDPSO, OLPSO, DPSO, and PSO.

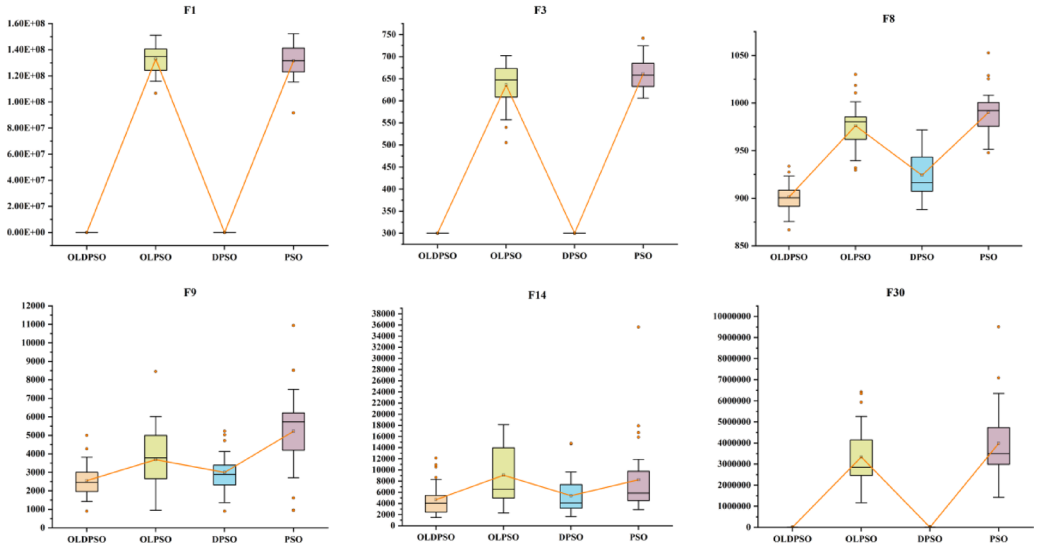


Fig. 8. Box plot comparison of OLDPSO, OLPSO, DPSO, and PSO performance.

Performance comparison of enhanced PSO algorithms

This section provides a comprehensive evaluation of OLDPSO's performance by comparing it with six improved PSO algorithms (ALCPSO, CGPSO, FST_PSO, ASCA_PSO, HCLPSO, and SCA_PSO) using the CEC 2017 benchmark functions, with results shown in Table 8. As indicated, OLDPSO outperforms all six variants, achieving the lowest ARV and ranking first overall. OLDPSO demonstrated particularly strong performance against FST_PSO and SCA_PSO, outperforming them on all 30 functions with a perfect R + score. Against CGPSO and ASCA_PSO, OLDPSO secured R + counts of 28 and 27, respectively, indicating superiority in nearly all functions. Furthermore, OLDPSO outperformed ALCPSO on 19 functions and HCLPSO on 16. In summary, OLDPSO exhibits outstanding performance compared to the six PSO variants, consistently leading across most test functions.

Table 9 presents the AvgR and StdR results for each algorithm across 30 test functions, providing a detailed comparison of their performance. The results highlight OLDPSO's consistent convergence toward the global optimum in most functions. Out of the 30 test functions, OLDPSO ranked in the top three for all except F23, F24, and F26, demonstrating high stability and performance. In contrast, FST_PSO achieved global optima in only a few test functions, with most of its results ranking in the middle range. OLDPSO's superior optimization capabilities over the six PSO variants are primarily due to the combined effect of the OL and D strategies. The OL mechanism significantly mitigates the likelihood of premature convergence, helping the algorithm avoid getting trapped in local optima at an early stage. Meanwhile, the D mechanism dynamically adjusts the swarm's control parameters, improving both global exploration and local exploitation, which enhances search efficiency and precision.

As shown in Fig. 10, OLDPSO consistently outperforms SCA_PSO, FST_PSO, CGPSO, and ASCA_PSO in almost all functions, with an R + score of 30 against SCA_PSO and FST_PSO, 28 against CGPSO, and 27 against ASCA_PSO. OLDPSO also surpasses ALCPSO in 19 out of 30 functions. The heatmap further confirms these results, highlighting OLDPSO's consistent performance across all functions, with HCLPSO being the most comparable in terms of results.

This study employs nine benchmark functions to evaluate the convergence behavior of various algorithms, as illustrated in Fig. 11. ALCPSO shows rapid convergence on F1, F7, F12, F19, and F30 functions. However, similar to most algorithms, it tends to converge prematurely to local optima. In contrast, OLDPSO effectively escapes local optima, progressively refining its solutions throughout the optimization process. For instance, in the unimodal function F2, OLDPSO initially converges at a similar rate to other algorithms. However, due to the integration of initialization with dynamic inertia weights and learning factors mechanisms, OLDPSO achieves superior final accuracy, effectively avoiding premature convergence and improving the precision of the solutions.

Figure 12 depicts box plots for six test functions, revealing that OLDPSO consistently exhibits the narrowest interquartile range among them. This suggests superior stability and robust optimization capabilities compared to other PSO variants.

Wall-clock time analysis of OLDPSO

We performed a comparative analysis of the execution times of OLDPSO, PSO, ALCPSO, CGPSO, FST_PSO, ASCA_PSO, HCLPSO, and SCA_PSO throughout the entire optimization process on CEC 2017 functions. The experiments were conducted using 30 search agents, with a maximum of 300,000 evaluations. The detailed wall-clock time consumed by each algorithm is reported in Table 10, measured in seconds. Furthermore, to provide a clearer visualization of the time consumption across different methods, a comparative wall-clock time chart is shown in Fig. 13.

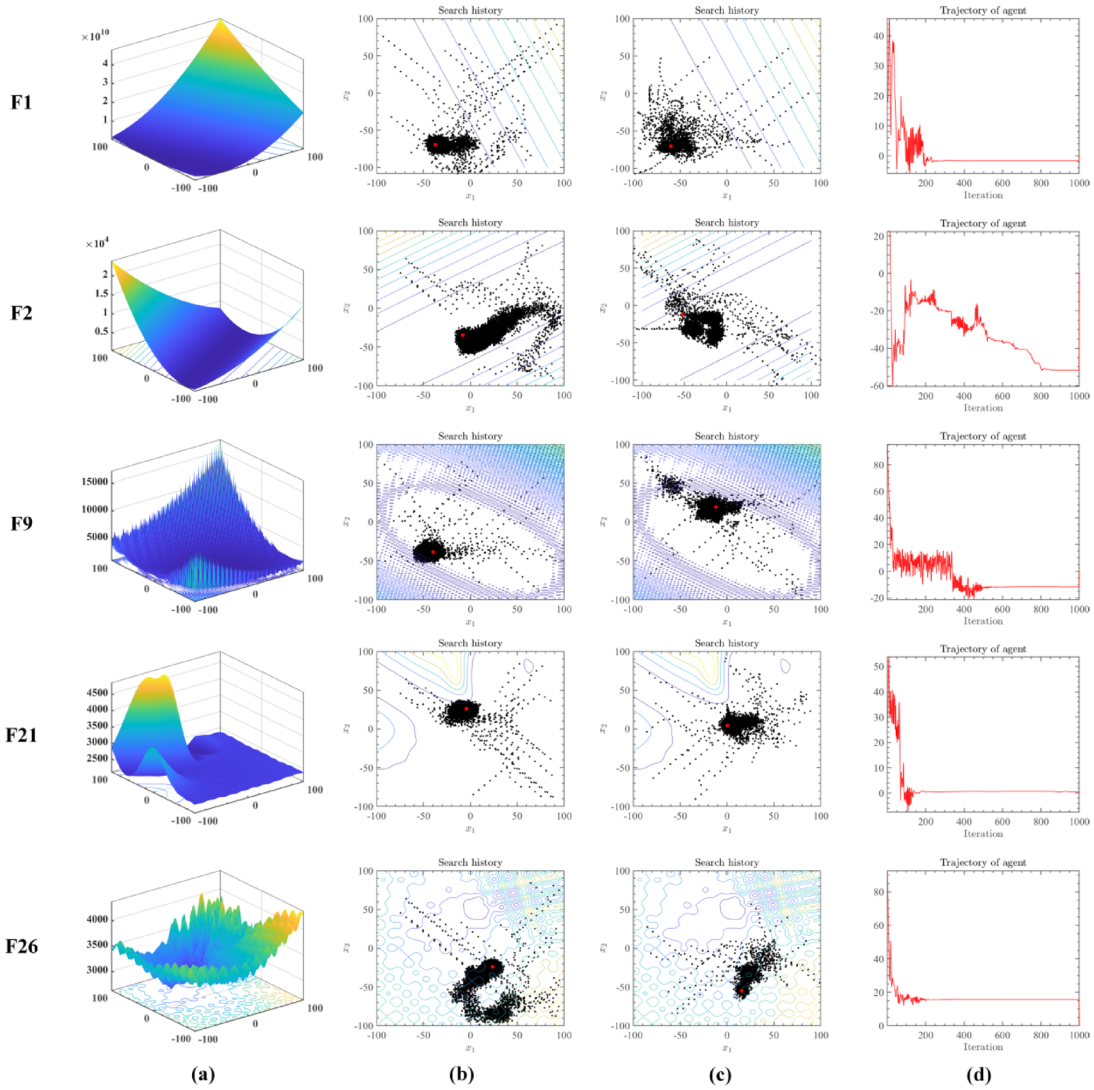


Fig. 9. Visual and qualitative comparison of OLDPSO and PSO performance: (a) 3D plots of benchmark functions, (b) Search trajectories of PSO, (c) Search trajectories of OLDPSO, and (d) Convergence curves of OLDPSO.

Algorithms	Rank	ARV	*R*+/*R*-/*R*=
OLDPSO	**1**	**2.0667**	–
ALCPSO	2	2.6667	19/11/0
CGPSO	4	4.4556	28/2/0
FST_PSO	7	6.3778	30/0/0
ASCA_PSO	5	4.7300	27/2/1
HCLPSO	3	2.8400	16/13/1
SCA_PSO	6	4.8633	30/0/0

Table 8. Rankings and performance metrics for OLDPSO and other enhanced PSO algorithms.

The experimental results indicate significant differences in the computational time consumed by different algorithms throughout the optimization process. Overall, ASCA_PSO and HCLPSO exhibit considerably longer execution times, with HCLPSO consuming substantially more time than other algorithms across multiple test functions, indicating its higher computational cost. In contrast, the other algorithms demonstrate relatively shorter execution times and perform similarly in most test functions. Notably, although OLDPSO integrates opposition-based Latin squares sampling initialization with dynamic inertia weights and learning factors to overcome the limitations of traditional PSO, its execution time remains comparable to that of the original PSO.

	AvgR	StdR	AvgR	StdR	AvgR	StdR
	F1		F2		F3	
OLDPSO	3.2025E + 03	6.5670E + 03	2.6329E + 07	1.0013E + 08	3.0000E + 02	2.7676E-04
ALCPSO	5.2464E + 03	5.4603E + 03	7.1652E + 18	2.2297E + 19	2.6050E + 04	2.3466E + 03
CGPSO	1.4207E + 08	1.7530E + 07	2.2310E + 14	4.0321E + 14	7.6280E + 02	6.2786E + 01
FST_PSO	2.2973E + 10	7.9615E + 09	1.1778E + 41	6.2703E + 41	8.8821E + 04	2.4897E + 04
ASCA_PSO	5.3514E + 08	1.0384E + 09	4.2035E + 30	1.6083E + 31	1.2455E + 03	3.4061E + 02
HCLPSO	3.0545E + 08	1.1717E + 08	5.1845E + 18	1.7432E + 19	5.5134E + 03	1.3369E + 03
SCA_PSO	3.4306E + 07	9.0815E + 06	2.1823E + 11	4.1578E + 11	4.2119E + 02	2.4528E + 01
	F4		F5		F6	
OLDPSO	4.6192E + 02	4.1616E + 01	6.4562E + 02	2.1835E + 01	6.2929E + 02	7.1895E + 00
ALCPSO	4.9920E + 02	2.6882E + 01	6.0202E + 02	3.4455E + 01	6.0562E + 02	5.4191E + 00
CGPSO	4.9140E + 02	3.0119E + 01	7.7510E + 02	4.0663E + 01	6.6072E + 02	1.0365E + 01
FST_PSO	4.0721E + 03	1.3998E + 03	8.0227E + 02	4.1107E + 01	6.6877E + 02	1.1132E + 01
ASCA_PSO	5.7854E + 02	1.2426E + 02	7.3299E + 02	3.5919E + 01	6.3487E + 02	8.2747E + 00
HCLPSO	4.9319E + 02	1.0408E + 01	6.3630E + 02	3.2818E + 01	6.0196E + 02	6.7165E-01
SCA_PSO	4.9021E + 02	3.6033E + 01	7.8352E + 02	4.7781E + 01	6.6757E + 02	7.6832E + 00
	F7		F8		F9	
OLDPSO	8.0471E + 02	2.2152E + 01	9.0003E + 02	1.7249E + 01	2.2508E + 03	7.8254E + 02
ALCPSO	8.5298E + 02	2.9128E + 01	8.9815E + 02	3.1269E + 01	1.7589E + 03	8.7222E + 02
CGPSO	9.2555E + 02	2.2839E + 01	1.0146E + 03	2.6590E + 01	6.7856E + 03	2.0561E + 03
FST_PSO	1.2975E + 03	9.3002E + 01	1.0567E + 03	3.6794E + 01	7.2204E + 03	1.7246E + 03
ASCA_PSO	9.9241E + 02	2.5389E + 01	1.0007E + 03	3.3412E + 01	4.8453E + 03	1.5916E + 03
HCLPSO	1.0920E + 03	2.0022E + 01	9.6793E + 02	2.4991E + 01	1.2873E + 03	1.3545E + 02
SCA_PSO	1.0212E + 03	3.2427E + 01	1.0102E + 03	3.0045E + 01	9.4205E + 03	1.8154E + 03
	F10		F11		F12	
OLDPSO	4.5309E + 03	7.1842E + 02	1.1943E + 03	2.6448E + 01	5.0565E + 04	5.9792E + 04
ALCPSO	4.3225E + 03	8.1598E + 02	1.2454E + 03	5.0759E + 01	5.7234E + 05	9.2917E + 05
CGPSO	6.2195E + 03	5.1651E + 02	1.2768E + 03	3.7790E + 01	2.6516E + 07	1.0379E + 07
FST_PSO	6.8261E + 03	6.7964E + 02	3.8391E + 03	1.7072E + 03	2.2451E + 09	1.0267E + 09
ASCA_PSO	6.1529E + 03	7.4940E + 02	1.3009E + 03	3.9811E + 01	5.3121E + 07	8.3892E + 07
HCLPSO	6.6725E + 03	7.0221E + 02	1.2447E + 03	3.3676E + 01	1.3355E + 06	1.0864E + 06
SCA_PSO	5.7267E + 03	6.9225E + 02	1.2963E + 03	3.9802E + 01	2.2180E + 07	1.3127E + 07
	F13		F14		F15	
OLDPSO	2.0300E + 04	1.5270E + 04	4.0235E + 03	2.3659E + 03	9.0178E + 03	7.3093E + 03
ALCPSO	2.2646E + 04	2.2949E + 04	4.0619E + 04	6.2252E + 04	1.7278E + 04	1.4917E + 04
CGPSO	5.2711E + 06	1.3524E + 06	1.0378E + 04	1.0121E + 04	5.3806E + 05	2.5895E + 05
FST_PSO	2.6398E + 08	4.3787E + 08	1.2735E + 05	2.9492E + 05	4.5756E + 04	3.8385E + 04
ASCA_PSO	1.0422E + 07	1.2625E + 07	2.7742E + 04	2.5367E + 04	1.2348E + 06	6.7948E + 05
HCLPSO	1.6541E + 04	2.1335E + 04	8.0477E + 03	3.9791E + 03	1.0836E + 04	1.9154E + 04
SCA_PSO	1.5661E + 06	8.7177E + 05	8.3640E + 03	4.4874E + 03	2.1718E + 05	1.0989E + 05
	F16		F17		F18	
OLDPSO	2.4985E + 03	2.4402E + 02	2.0615E + 03	1.6015E + 02	7.9890E + 04	4.6501E + 04
ALCPSO	2.5652E + 03	2.7277E + 02	2.1516E + 03	1.7422E + 02	2.1366E + 05	1.7414E + 05
CGPSO	2.9830E + 03	2.3415E + 02	2.3458E + 03	1.8292E + 02	1.9974E + 05	8.0503E + 04
FST_PSO	3.9382E + 03	5.6283E + 02	2.6067E + 03	2.9959E + 02	2.5407E + 06	3.8758E + 06
ASCA_PSO	2.9779E + 03	2.7733E + 02	2.2834E + 03	1.6642E + 02	5.0017E + 05	3.4440E + 05
HCLPSO	2.2064E + 03	1.9396E + 02	1.8185E + 03	4.4935E + 01	3.6186E + 05	1.6388E + 05
SCA_PSO	3.4164E + 03	3.5775E + 02	2.6377E + 03	3.0513E + 02	1.3474E + 05	1.0053E + 05
	F19		F20		F21	
OLDPSO	9.8032E + 03	9.4260E + 03	2.3993E + 03	1.5079E + 02	2.4181E + 03	2.3249E + 01
ALCPSO	1.7905E + 04	1.6933E + 04	2.3563E + 03	1.4791E + 02	2.4146E + 03	3.2388E + 01
CGPSO	1.3572E + 06	8.5682E + 05	2.6265E + 03	1.9141E + 02	2.5470E + 03	3.4025E + 01
FST_PSO	6.9446E + 06	1.1448E + 07	2.8684E + 03	2.1234E + 02	2.6052E + 03	3.7996E + 01
ASCA_PSO	4.5552E + 06	2.1130E + 06	2.5249E + 03	1.3265E + 02	2.4836E + 03	2.9352E + 01
HCLPSO	1.4810E + 04	1.7533E + 04	2.1854E + 03	8.3143E + 01	2.4146E + 03	4.1367E + 01
SCA_PSO	2.6601E + 06	1.3890E + 06	2.7504E + 03	1.8931E + 02	2.5848E + 03	5.5560E + 01

Continued

	AvgR	StdR	AvgR	StdR	AvgR	StdR
	F22		F23		F24	
OLDPSO	4.1237E + 03	2.0148E + 03	2.9655E + 03	1.6582E + 02	3.0862E + 03	6.2380E + 01
ALCPSO	4.3887E + 03	1.6959E + 03	2.7919E + 03	4.7134E + 01	2.9720E + 03	5.5795E + 01
CGPSO	4.1214E + 03	2.5536E + 03	3.1231E + 03	1.2629E + 02	3.2089E + 03	8.6405E + 01
FST_PSO	7.7124E + 03	1.4210E + 03	3.3924E + 03	1.9462E + 02	3.4165E + 03	1.6795E + 02
ASCA_PSO	6.1169E + 03	2.1929E + 03	2.8802E + 03	4.1083E + 01	3.0235E + 03	4.0251E + 01
HCLPSO	2.3277E + 03	5.3365E + 00	2.6981E + 03	1.6729E + 01	2.8790E + 03	2.0891E + 01
SCA_PSO	6.8046E + 03	1.7275E + 03	3.4312E + 03	1.4041E + 02	3.4554E + 03	1.1009E + 02
	F25		F26		F27	
OLDPSO	2.8812E + 03	6.0779E + 00	5.8563E + 03	1.8026E + 03	3.1746E + 03	1.4657E + 02
ALCPSO	2.8985E + 03	1.5106E + 01	4.6236E + 03	7.9931E + 02	3.2630E + 03	4.2993E + 01
CGPSO	2.9196E + 03	2.5748E + 01	4.8051E + 03	2.1127E + 03	3.2207E + 03	1.4637E + 02
FST_PSO	3.7338E + 03	3.3516E + 02	8.7821E + 03	9.9792E + 02	3.6972E + 03	2.7321E + 02
ASCA_PSO	2.9283E + 03	4.2522E + 01	6.3258E + 03	4.6276E + 02	3.3066E + 03	3.3840E + 01
HCLPSO	2.9034E + 03	1.1109E + 01	3.3606E + 03	2.7330E + 02	3.2049E + 03	5.5839E + 00
SCA_PSO	2.9056E + 03	2.2743E + 01	8.4132E + 03	2.7127E + 03	3.9077E + 03	4.9494E + 02
	F28		F29		F30	
OLDPSO	3.2137E + 03	2.8343E + 01	3.6098E + 03	1.9977E + 02	6.1704E + 03	3.9081E + 03
ALCPSO	3.2331E + 03	4.1258E + 01	3.8503E + 03	2.3686E + 02	1.4595E + 04	6.8587E + 03
CGPSO	3.2586E + 03	2.3073E + 01	4.4026E + 03	2.4671E + 02	4.0182E + 06	1.4233E + 06
FST_PSO	4.6235E + 03	6.3800E + 02	5.5731E + 03	5.8005E + 02	2.9785E + 07	3.3474E + 07
ASCA_PSO	3.3286E + 03	1.1193E + 02	4.2820E + 03	2.6506E + 02	8.3218E + 06	4.4318E + 06
HCLPSO	3.2373E + 03	1.7495E + 01	3.4988E + 03	9.8099E + 01	5.0301E + 04	1.0809E + 05
SCA_PSO	3.2686E + 03	1.9475E + 01	4.7574E + 03	3.4202E + 02	5.2547E + 06	3.0637E + 06

Table 9. Comparative AvgR and StdR of OLDPSO and other enhanced PSO algorithms.

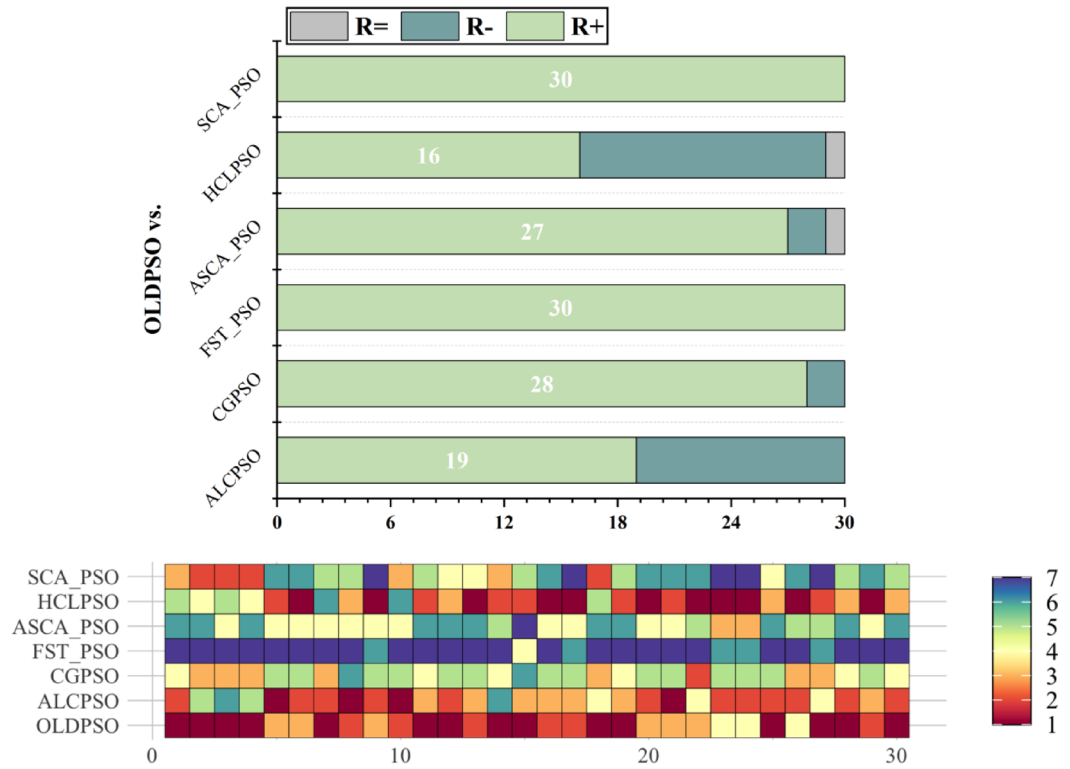


Fig. 10. Performance comparison of OLDPSO and other enhanced PSO variants.

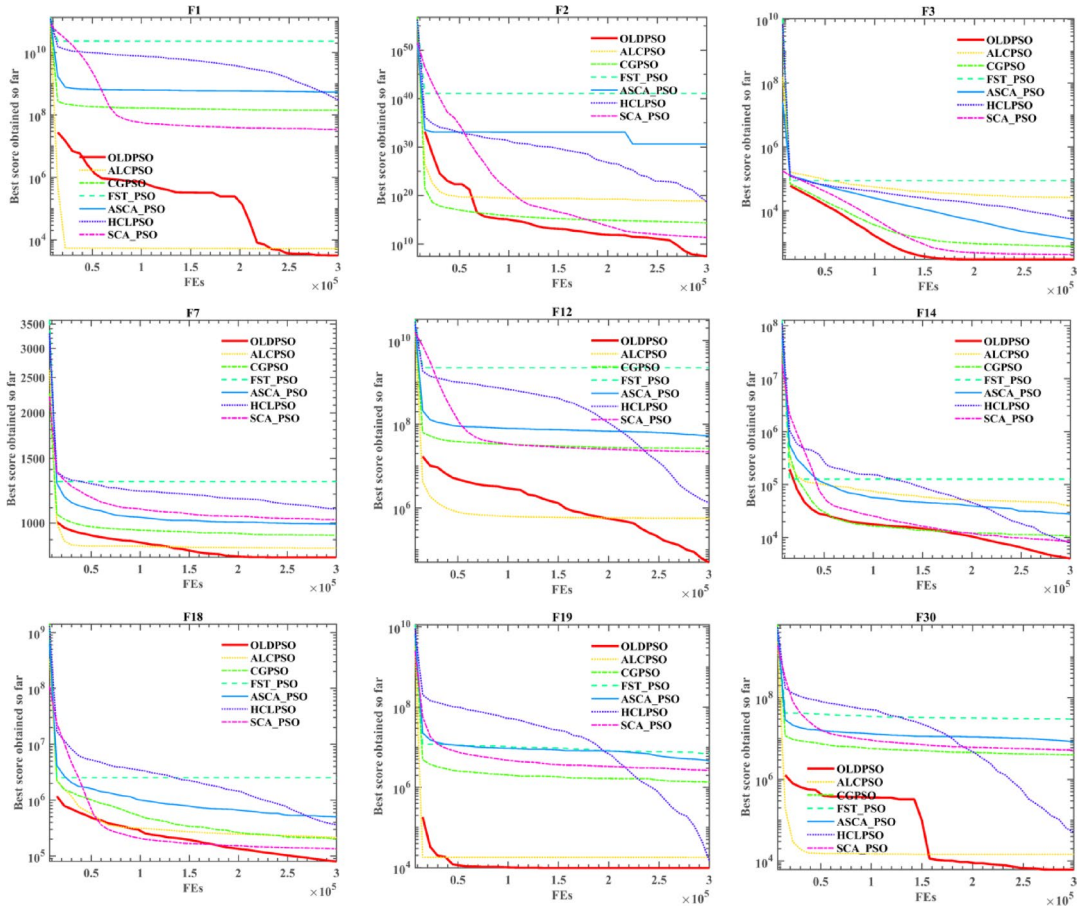


Fig. 11. Convergence performance comparison of OLDPSO and other enhanced PSO variants.

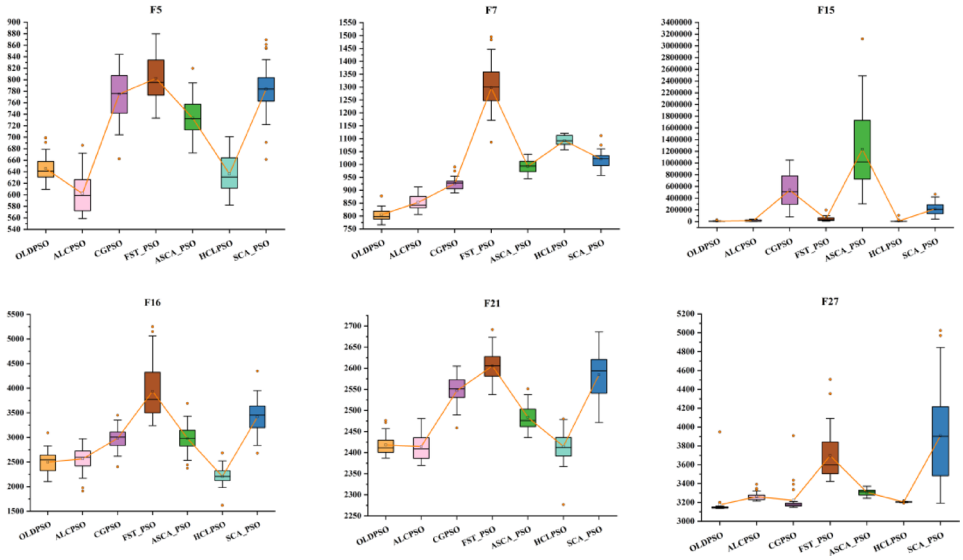


Fig. 12. Box plot comparison of OLDPSO and other enhanced PSO variants.

Experimental applications of OLDPSO-SVM to AD diagnosis

This paper focuses on optimizing the performance of the OLDPSO-SVM algorithm for AD diagnosis. The diagnostic tasks involve distinguishing between AD, NC, and MCI, specifically in AD vs. NC, NC vs. MCI-NC, NC vs. MCI-C, MCI-NC vs. MCI-C, MCI-NC vs. AD, and MCI-C vs. AD. First, different TFs are compared

F	OLDPSO	PSO	ALCPSO	CGPSO	FST_PSO	ASCA_PSO	HCLPSO	SCA_PSO
F1	5.63	5.60	5.75	5.20	5.72	8.15	11.21	5.89
F2	5.60	5.48	5.66	5.17	5.62	8.29	12.31	5.88
F3	5.58	5.52	5.77	5.16	5.60	8.26	12.35	5.87
F4	5.53	5.54	5.74	5.17	5.66	8.15	15.31	5.88
F5	5.78	5.83	5.88	5.44	5.79	8.54	12.90	6.13
F6	7.49	7.48	7.55	7.10	7.39	10.10	14.85	7.86
F7	6.07	6.01	6.25	5.65	5.96	8.78	9.87	6.46
F8	5.97	6.11	6.11	5.56	5.85	8.76	12.75	6.33
F9	5.98	6.00	5.95	5.61	5.95	8.66	12.66	6.32
F10	6.73	6.86	7.06	6.41	6.73	9.54	15.13	7.10
F11	5.89	5.86	6.12	5.48	5.88	8.52	12.65	6.21
F12	6.31	6.30	6.45	5.93	6.35	8.97	14.17	6.50
F13	6.18	6.14	6.34	5.74	6.12	8.74	13.84	6.45
F14	6.52	6.59	6.78	6.11	6.48	9.16	14.20	6.66
F15	6.11	6.10	6.32	5.67	6.09	8.81	13.95	6.38
F16	6.40	6.36	6.49	5.96	6.27	9.11	14.35	6.66
F17	6.86	6.79	6.93	6.44	6.76	9.62	14.98	7.13
F18	6.22	6.22	6.40	5.79	6.16	9.00	14.28	6.65
F19	10.91	11.17	11.16	10.64	10.26	14.01	18.77	11.19
F20	7.34	7.26	7.49	6.79	7.14	9.90	14.72	7.52
F21	9.13	9.12	9.34	8.70	8.95	12.24	17.71	9.36
F22	10.34	10.32	10.38	9.73	9.84	13.10	17.47	10.43
F23	10.85	10.95	11.00	10.50	10.67	14.02	22.13	11.19
F24	10.55	10.71	11.07	10.26	10.33	13.75	21.97	10.95
F25	10.82	10.77	11.18	10.34	10.45	13.93	18.49	11.11
F26	12.06	12.28	12.53	11.79	11.77	15.18	20.66	12.46
F27	13.49	13.27	13.75	12.84	12.93	16.20	25.76	13.54
F28	12.28	12.26	12.60	11.76	11.79	15.09	20.32	12.52
F29	10.17	10.23	10.50	9.69	9.86	13.19	19.26	10.35
F30	14.17	14.12	14.57	13.62	13.45	17.09	22.97	14.57

Table 10. Detailed wall-clock time consumption of each algorithm.

to enhance classification accuracy and stability. Next, the performance of GM, WM, and their combination is evaluated across various diagnostic tasks. Following that, OLDPSO-SVM is compared with other machine learning and deep learning models, highlighting its superior performance. Finally, GM and WM feature selection results are presented, identifying key brain regions for classification.

Performance comparison of different TFs in OLDPSO-SVM

The objective of this experiment, which was based on the combined use of GM and WM data, was to identify the most suitable TF for different classification tasks to optimize the performance of the OLDPSO-SVM algorithm. Specifically, the objective was to compare the performance of various TFs in terms of accuracy AvgR and StdR, and determine the optimal TF for each classification task.

As shown in Table 11, in the AD vs. NC classification task, the V3 performed the best, achieving an accuracy of 0.99111 with a StdR of 0.018754, indicating both high accuracy and remarkable stability. For the NC vs. MCI-NC task, the S1 was the most effective, with an accuracy of 0.8976. Although the S3 had a slightly lower StdR (S3: 0.047054 vs. S1: 0.049048), S1's higher accuracy made it the superior choice. In the NC vs. MCI-C task, the V2 excelled, achieving an accuracy of 0.99069 and a StdR of 0.019628, showcasing both excellent classification performance and robustness.

For more challenging tasks, such as MCI-NC vs. MCI-C, the S1 emerged as the best performer, with an accuracy of 0.88382. However, its relatively high StdR (0.079168) suggests greater variability in classification outcomes. In the MCI-NC vs. AD task, the S1 again showed strong performance, achieving an accuracy of 0.94688 and a StdR of 0.035789, demonstrating both high accuracy and stability in this task. Lastly, in the MCI-C vs. AD task, the V2 delivered the best performance, with an accuracy of 0.87956 and a StdR of 0.038667. Despite the inherent difficulty of this classification task due to overlapping features between MCI-C and AD, the V2 effectively captured the subtle distinctions between the two groups, leading to reliable performance.

Figure 14 provides a clear visualization of the performance of different TFs across multiple classification tasks, combining both a heatmap and line plot with error bands to depict the accuracy and uncertainty of each TF. The heatmap highlights differences in accuracy, with darker shades indicating higher classification performance, while the line plot further reveals the trends and stability of the TFs across different tasks. Compared to Table 11,



Fig. 13. Comparison of wall-clock times for different algorithms.

TFs	AvgR	StdR	AvgR	StdR	AvgR	StdR
	AD vs. NC		NC vs. MCI-NC		NC vs. MCI-C	
S1	0.98613	0.022364	**0.8976**	0.049048	0.97143	0.046004
S2	0.99048	0.020078	0.88485	0.053106	0.97141	0.033327
S3	0.99089	0.01923	0.87564	**0.047054**	0.96119	0.038458
S4	0.98117	0.033096	0.84762	0.061027	0.97095	0.040419
V1	0.98634	0.022009	0.88896	0.069039	0.97571	0.025614
V2	0.98593	0.022661	0.87482	0.077413	**0.99069**	**0.019628**
V3	**0.99111**	**0.018754**	0.86126	0.048027	0.97071	0.025224
V4	0.98615	0.022313	0.88849	0.051326	0.97071	0.033766
	MCI-NC vs. MCI-C		MCI-NC vs. AD		MCI-C vs. AD	
S1	**0.88382**	0.079168	**0.94688**	**0.035789**	0.87298	0.057386
S2	0.87447	**0.042179**	0.91844	0.044064	0.84858	0.070589
S3	0.83351	0.085791	0.90381	0.049718	0.84261	0.070556
S4	0.82375	0.055692	0.91258	0.050356	0.83877	0.050725
V1	0.85456	0.047349	0.92805	0.052263	0.86887	0.048413
V2	0.84896	0.080523	0.93184	0.053653	**0.87956**	**0.038667**
V3	0.8452	0.070041	0.91353	0.054043	0.85396	0.04875
V4	0.83909	0.080688	0.92303	0.064494	0.85361	0.078378

Table 11. Accuracy performance of different TFs with the OLDPSO-SVM algorithm. Bold values indicate the highest AvgR and the lowest StdR achieved for each classification task, highlighting the optimal performance in terms of both effectiveness and stability

the error bands in the line plot offer a clearer representation of the variability in classification outcomes. This is particularly evident in more challenging tasks, such as NC vs. MCI-NC and MCI-NC vs. MCI-C, where the performance of the TFs shows greater fluctuation. In contrast, the AD vs. NC task demonstrates consistently high accuracy with minimal variability.

Figure 15 illustrates the fitness convergence of the OLDPSO-SVM algorithm across six different classification tasks, using eight different TFs. The objective is to identify the optimal TF for each classification task. The horizontal axis indicates the number of algorithm iterations, while the vertical axis shows the average fitness values obtained after ten independent runs. The results indicate that V3 performs best in the AD vs. NC task, with the fastest convergence and the lowest fitness value. In the NC vs. MCI-NC, MCI-NC vs. MCI-C, and MCI-NC vs. AD tasks, S1 demonstrates the lowest fitness value, making it the most effective TF. For the NC vs. MCI-C and MCI-C vs. AD tasks, V2 achieves the fastest convergence and the lowest fitness value, making it the optimal TF in these cases. In subsequent experiments, we will adopt the most effective TF identified for each classification task.

Performance evaluation of GM, WM, and combined models

This study aims to assess the classification performance of GM and WM in distinguishing between AD, NC, and individuals with MCI. The study compares the effectiveness of using GM, WM, and their combination (GM + WM) across various classification tasks. As shown in Table 12, the GM + WM combined model demonstrates superior performance in all classification tasks.

In the classification between AD and NC, the GM + WM model achieved an accuracy of 99.11%±1.88%, with sensitivity and specificity around 99%, outperforming the GM (98.10%±3.33%) and WM (93.96%±6.19%) models. For NC vs. MCI-NC, the GM + WM model also performed best, with 89.76%±4.90% accuracy, 91.82%±10.00% sensitivity, and 87.36%±12.62% specificity, surpassing the individual GM and WM models. In NC vs. MCI-C, the GM + WM model achieved an accuracy of 99.07%±1.96%, again outperforming the GM and WM models. For the more challenging MCI-NC vs. MCI-C task, GM + WM achieved 88.38% ± 7.92% accuracy, delivering the best performance despite the subtle differences between the groups. In MCI-NC vs. AD, the GM + WM model showed strong results, with 94.69% ± 3.58% accuracy, outperforming the GM and WM models. Finally, for MCI-C vs. AD, the GM + WM model achieved 87.96% ± 3.87% accuracy, again surpassing the individual models.

In Fig. 16, the mean value and standard deviation for various classification metrics across different classification tasks (AD vs. NC, NC vs. MCI-C, and MCI-NC vs. AD) are presented. The results demonstrate that the combined GM and WM model (GM + WM) consistently outperforms the individual GM and WM models, with mean values approaching 1 in key metrics such as accuracy, specificity, Precision, and F-measure, indicating superior classification performance. Moreover, the standard deviation results highlight the stability and consistency of the GM + WM model. In several metrics, the GM + WM model exhibits significantly lower standard deviations compared to the individual GM and WM models, particularly in MCC and F-measure, indicating less variability and more consistent classification outcomes. In contrast, the individual GM and WM models display larger standard deviations, especially in Sensitivity and other metrics, with the WM model showing particularly high variability, indicating less stable performance across different experiments.

Overall, the GM + WM model not only achieves excellent performance in terms of classification metrics, but its lower standard deviations suggest higher robustness, with more stable and reliable results across experiments. These results suggest that combining GM and WM provides a more comprehensive representation of brain pathology, leading to significantly improved accuracy in disease classification.

Comparison with classical machine learning algorithms

The objective of this experiment is to evaluate the performance of the proposed OLDPSO-SVM algorithm in comparison with other commonly used machine learning models, specifically Back Propagation Neural Network (BP), Classification and Regression Tree (CART), Random Forest (RandomF), AdaBoost, and Extreme Learning Machine for Feature Selection (ELMforFS), for classifying individuals with AD, MCI, and NC. The aim is to identify a more effective method for early diagnosis. As shown in Fig. 17, the results demonstrate that the proposed OLDPSO-SVM algorithm outperforms these baseline models across various classification tasks, particularly in key metrics such as accuracy, sensitivity, specificity, precision, MCC, and F-measure. OLDPSO-SVM exhibited superior stability and classification capability in addressing AD-related classification problems, highlighting its potential for significant application in early diagnostic processes.

Figure 18 presents the accuracy performance of various models across each fold in a 10-fold cross-validation setting for different classification tasks. Each color and position in the bars represent a distinct machine learning model. The figure reveals that the OLDPSO-SVM model maintains a consistently high and stable accuracy across the majority of folds in each task, significantly outperforming the other machine learning models. In contrast, the accuracy of the other models exhibits greater variability, particularly in more complex tasks, such as MCI-NC vs. MCI-C and MCI-C vs. AD, where the performance differences between models become more pronounced. Overall, the results indicate that OLDPSO-SVM demonstrates robust classification capability and reliability in handling AD-related classification tasks.

Comparison with other existing methods

To evaluate the performance of the proposed OLDPSO-SVM in AD recognition, we conducted a comparative analysis against previous deep learning methods, such as CNNs and Transformers, on the ADNI dataset, as summarized in Table 13. It is important to note that due to differences in data processing protocols across studies, this comparison serves as a reference. Despite these variations, our results demonstrate that the proposed OLDPSO-SVM achieved a satisfactory accuracy, second only to Liu et al.[56] (99.4%) and outperforming

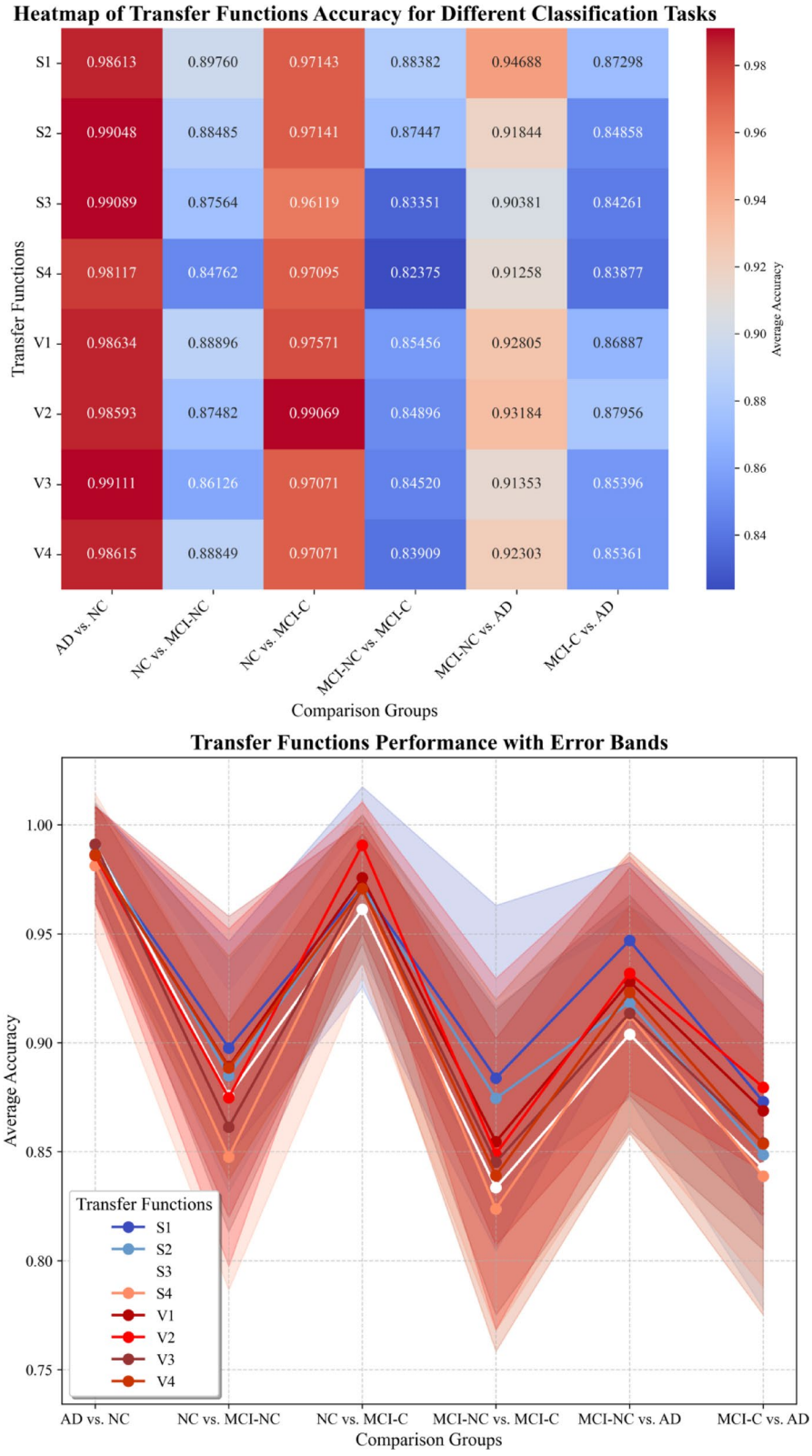


Fig. 14. Performance comparison of TFs across multiple classification tasks.

most existing methods reported in the literature. Although CNNs and Transformers have advantages in end-to-end feature learning, the core strength of OLDPSO-SVM lies in its unique feature selection capabilities and interpretability. By precisely identifying key brain regions associated with AD, this model can provide biologically meaningful insights for medical diagnoses. Furthermore, it demonstrates robust classification performance on

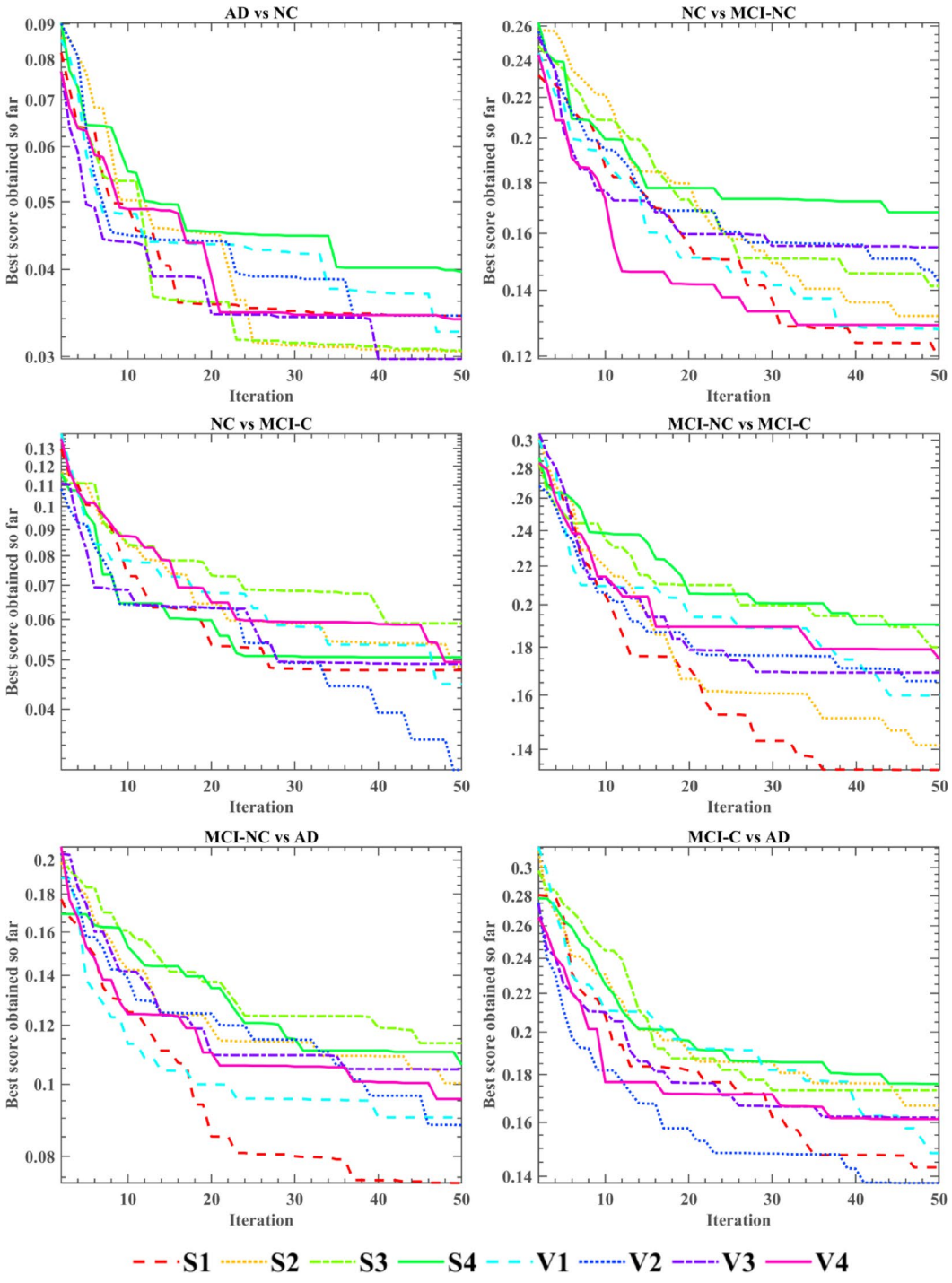


Fig. 15. Fitness convergence of OLDPSO-SVM algorithm using different TFs across six classification tasks.

small sample datasets, significantly outperforming deep learning models that rely on large-scale data training, such as the CNN developed by Yagis et al.[50], which achieved only 73.4% accuracy. This broader comparison further underscores the effectiveness and potential advantages of OLDPSO-SVM in AD recognition.

GM and WM feature selection for AD diagnosis using OLDPSO-SVM

Figure 19 presents the GM and WM feature selection results across AD, MCI, and NC groups, derived from the OLDPSO-SVM algorithm using 10-fold cross-validation, repeated 10 times. The vertical axis represents the total number of times each brain region was selected, while the horizontal axis (R1 to R90) corresponds to the 90 regions of interest (ROIs) defined by the AAL template, as described in Table 2. These feature selection results underscore the importance of specific brain regions in the classification tasks, with regions frequently selected indicating their crucial role in distinguishing between AD, MCI, and NC groups. To offer a more intuitive

	Accuracy	Sensitivity	Specificity
	AD vs. NC		
GM	98.095%±3.3296%	97%±6.7495%	99.091%±2.8748%
WM	93.961%±6.1944%	92.182%±8.9402%	95.53%±6.3729%
GM + WM	99.111%±1.8754%	99.091%±2.8748%	99.167%±2.6352%
	Precision	**MCC**	**F-measure**
	AD vs. NC		
GM	99.091%±2.8748%	96.401%±6.2486%	97.886%±3.7883%
WM	95.328%±6.5648%	88.202%±12.327%	93.528%±6.5105%
GM + WM	99.167%±2.6352%	98.295%±3.5948%	99.089%±1.9230%
	Accuracy	**Sensitivity**	**Specificity**
	NC vs. MCI-NC		
GM	87.035%±4.2994%	90%±7.96%	83.727%±11.253%
WM	88.411%±4.0309%	91.136%±8.0496%	85.545%±10.369%
GM + WM	89.76%±4.9048%	91.818%±10.005%	87.364%±12.622%
	Precision	**MCC**	**F-measure**
	NC vs. MCI-NC		
GM	86.714%±8.4465%	75.04%±8.413%	87.785%±4.0146%
WM	88.136%±8.0178%	77.8%±7.9388%	89.074%±3.5852%
GM + WM	90.079%±8.0983%	80.801%±8.6465%	90.255%±4.5094%
	Accuracy	**Sensitivity**	**Specificity**
	NC vs. MCI-C		
GM	95.714%±5.2405%	98.182%±5.7496%	92.889%±10.634%
WM	89.377%±4.4103%	88.333%±8.5128%	90.667%±5.7664%
GM + WM	99.069%±1.9628%	99.167%±2.6352%	98.889%±3.5136%
	Precision	**MCC**	**F-measure**
	NC vs. MCI-C		
GM	94.716%±7.7838%	92.042%±9.5743%	96.132%±4.699%
WM	91.9%±4.5798%	79.264%±8.662%	89.826%±4.7662%
GM + WM	99.231%±2.4325%	98.187%±3.8259%	99.165%±1.7618%
	Accuracy	**Sensitivity**	**Specificity**
	MCI-NC vs. MCI-C		
GM	86.377%±5.5273%	88.455%±7.7856%	84.333%±12.356%
WM	85.808%±5.9773%	86.364%±9.64%	85.444%±9.4201%
GM + WM	88.382%±7.9168%	89.364%±9.8011%	87.222%±12.992%
	Precision	**MCC**	**F-measure**
	MCI-NC vs. MCI-C		
GM	87.147%±9.9344%	73.763%±11.453%	87.241%±4.8288%
WM	87.052%±7.494%	72.381%±11.441%	86.272%±6.1741%
GM + WM	89.562%±9.9702%	77.514%±15.449%	89.016%±7.641%
	Accuracy	**Sensitivity**	**Specificity**
	MCI-NC vs. AD		
GM	91.396%±6.4769%	89.636%±13.3%	93.091%±9.4592%
WM	90.333%±3.9606%	94.182%±6.8862%	86.273%±9.5965%
GM + WM	94.688%±3.5789%	94.455%±6.4204%	95.182%±6.6536%
	Precision	**MCC**	**F-measure**
	MCI-NC vs. AD		
GM	93.928%±7.297%	84.019%±11.715%	90.941%±7.3702%
WM	88.262%±7.3766%	81.582%±7.6282%	90.758%±3.8005%
GM + WM	95.515%±6.1638%	89.916%±6.7823%	94.71%±3.5929%
	Accuracy	**Sensitivity**	**Specificity**
	MCI-C vs. AD		
GM	85.88%±6.1528%	83.222%±11.005%	88.545%±10.775%
WM	84.406%±6.1276%	79%±13.673%	89.364%±9.6023%
GM + WM	87.956%±3.8667%	87.222%±6.8743%	88.455%±8.6674%
	Precision	**MCC**	**F-measure**
	MCI-C vs. AD		

Continued

	Precision	MCC	F-measure
GM	88.275%±10.41%	72.936%±11.846%	84.838%±6.5572%
WM	88.662%±10.037%	70.125%±11.764%	82.477%±7.7329%
GM + WM	88.51%±7.7635%	76.462%±7.6242%	87.436%±3.5762%

Table 12. Performance comparison of GM and WM in distinguishing AD, NC, and MCI.

depiction of the results shown in Figs. 19 and 20 provides 3D visualizations of the top five most frequently selected brain regions for each classification task. These 3D brain maps vividly illustrate the structural differences in both GM and WM across the AD, MCI, and NC groups.

In the AD vs. NC classification task, the top five GM regions selected were the left and right hippocampus (R37, R38), the left supplementary motor area (R19), the left middle frontal gyrus (R8), the left and right inferior temporal gyrus (R89, R90), and the right heschl gyrus (R80). The top five WM regions selected were the left parahippocampal gyrus (R39), the right inferior occipital gyrus (R54), the left insula (R29), the right pallidum (R76), the left precentral gyrus (R1), and the right amygdala (R42).

In the NC vs. MCI-C classification task, the top five GM regions selected were the left and right hippocampus (R37, R38), the left olfactory (R21), the right temporal pole (R88), the left middle temporal gyrus (R85), and the left supramarginal gyrus (R63). The top five WM regions selected were the left parahippocampal gyrus (R39), the left medial superior frontal gyrus (R23), the left precuneus (R67), the right medial orbitofrontal cortex (R26), the left posterior cingulate gyrus (R35), and the left middle temporal pole (R87).

In the MCI-NC vs. MCI-C classification task, the top five GM regions selected were the right heschl gyrus (R80), the right middle temporal gyrus (R86), the left middle occipital gyrus (R51), the right paracentral lobule (R70), the left fusiform gyrus (R55), the left pallidum (R75), the right thalamus (R78), the right precentral gyrus (R2), and the right inferior occipital gyrus (R54). The top five WM regions selected were the right putamen (R74), the left posterior cingulate gyrus (R35), the left superior occipital gyrus (R49), the right middle temporal pole (R88), the right anterior cingulate gyrus (R32), and the left inferior temporal (R89).

This detailed selection of GM and WM regions highlights the key structural changes in the brain across the different classification tasks, emphasizing both hemispheric and regional involvement in the progression from NC to MCI and AD.

Discussion

This study optimized the performance of the OLDPSO-SVM algorithm across various classification tasks by comparing the effectiveness of different TFs. The results indicate significant differences in the applicability of the TFs across tasks, which can be analyzed from the perspectives of task complexity, inter-group structural differences, and algorithm adaptability. In simpler classification tasks (such as AD vs. NC), most TFs exhibited high accuracy and low StdR, indicating that the structural differences between groups were more distinct and easier to capture. The V3 performed the best, showing a faster and more efficient identification of key features differentiating the AD and NC groups. These types of tasks were less sensitive to the choice of TF. For more complex classification tasks (such as MCI-NC vs. MCI-C), although the S1 performed well in terms of accuracy, its higher StdR indicated greater variability in the results, highlighting the challenges in feature separation. Due to the significant structural overlap between the MCI-NC and MCI-C groups, classification boundaries are harder to define. While the S1 captured subtle differences, there is still room for improvement in terms of stability. In tasks such as NC vs. MCI-C and MCI-C vs. AD, the V2 demonstrated faster convergence and the lowest fitness values, indicating its advantage in handling complex tasks where feature boundaries are less distinct. V2 effectively differentiated between MCI-C and AD, showing potential in tracking the progression from MCI to AD.

Overall, the higher StdR observed in complex tasks suggests that current TFs exhibit some instability when dealing with highly overlapping groups. Future research should focus on developing more stable TFs and integrating multimodal data to improve classification accuracy and robustness. Additionally, the convergence advantages of S1 and V2 in complex tasks suggest their potential for reducing errors in decision boundaries. Future studies should further optimize their parameter configurations or incorporate adaptive mechanisms.

The feature selection results for GM and WM across AD, MCI, and NC groups provide valuable insights into the structural differences in the brain associated with these conditions. The application of the OLDPSO-SVM algorithm, combined with 10-fold cross-validation repeated 10 times, ensures a robust identification of the most relevant brain regions for classification tasks. This analysis highlights the significance of certain brain regions in differentiating between these groups, indicating their potential as biomarkers for early diagnosis and distinguishing AD from other cognitive impairments. In the AD vs. NC classification task, the hippocampus emerged as a key GM region. The hippocampus, which plays a crucial role in memory formation, is among the first regions affected by AD, thus explaining its frequent selection. Additionally, the involvement of other regions, such as the middle frontal gyrus and the inferior temporal gyrus, emphasizes the widespread cortical degeneration associated with AD. The selected WM regions, including the parahippocampal gyrus and the insula, are consistent with previous studies that highlight the disruption of WM integrity as a hallmark of AD progression. For the NC vs. MCI classification task, a similar pattern of hippocampus involvement in GM was observed, reinforcing this region's critical role in the early stages of cognitive decline. Interestingly, the frequent selection of regions like the olfactory cortex and the supramarginal gyrus suggests that the olfactory system, often overlooked in neurodegenerative studies, may play a more prominent role in the early detection of cognitive

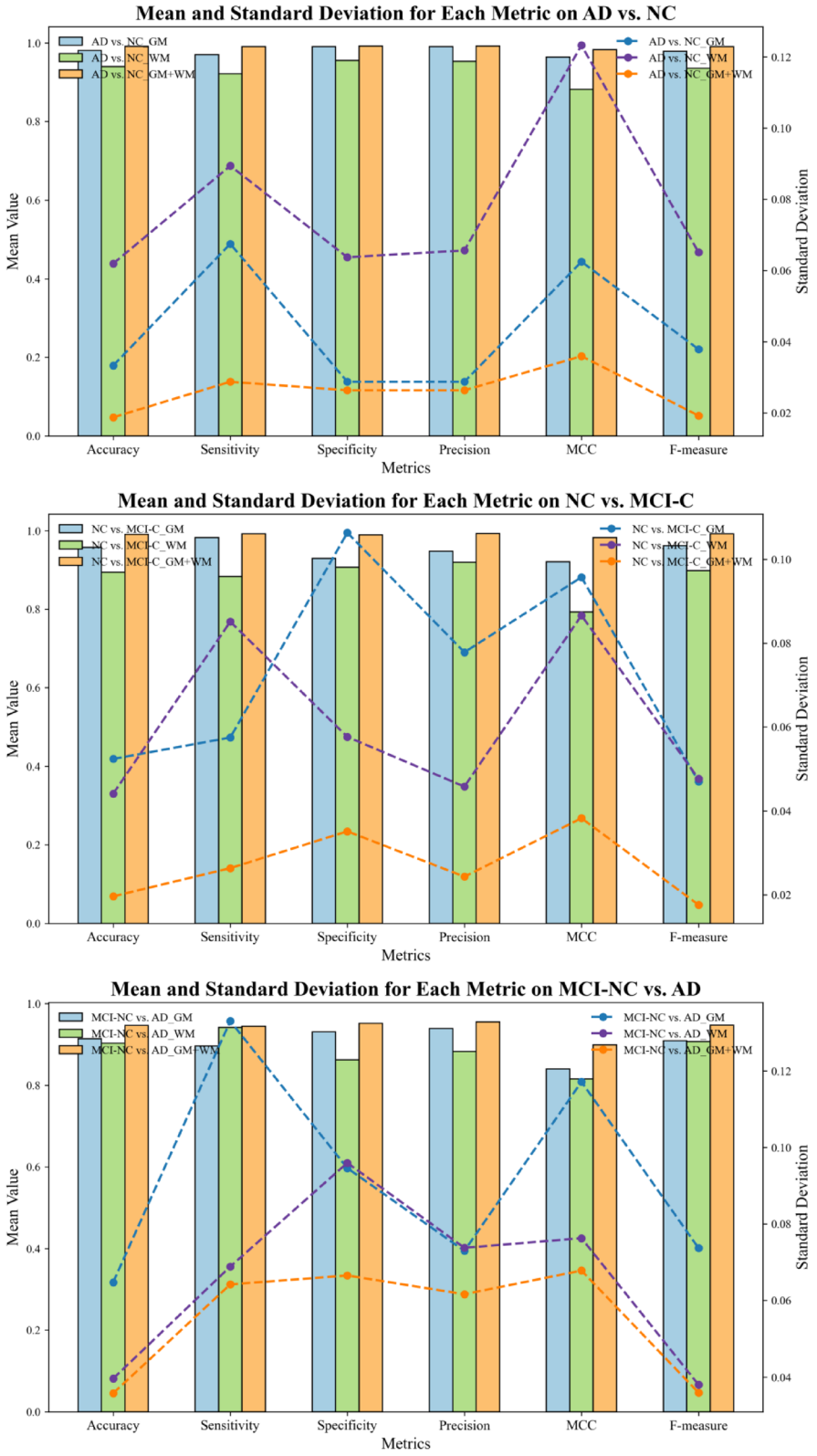


Fig. 16. Mean and standard deviation of classification metrics for GM, WM, and GM + WM models in distinguishing AD, NC, and MCI.

decline than previously thought. The WM regions selected in this task, such as the posterior cingulate gyrus and the orbitofrontal cortex, are also heavily implicated in cognitive processing and memory, further validating their relevance to MCI classification. In the MCI-NC vs. MCI-C task, which is particularly challenging due to the subtle structural differences between non-converting and converting MCI, GM regions such as the heschl

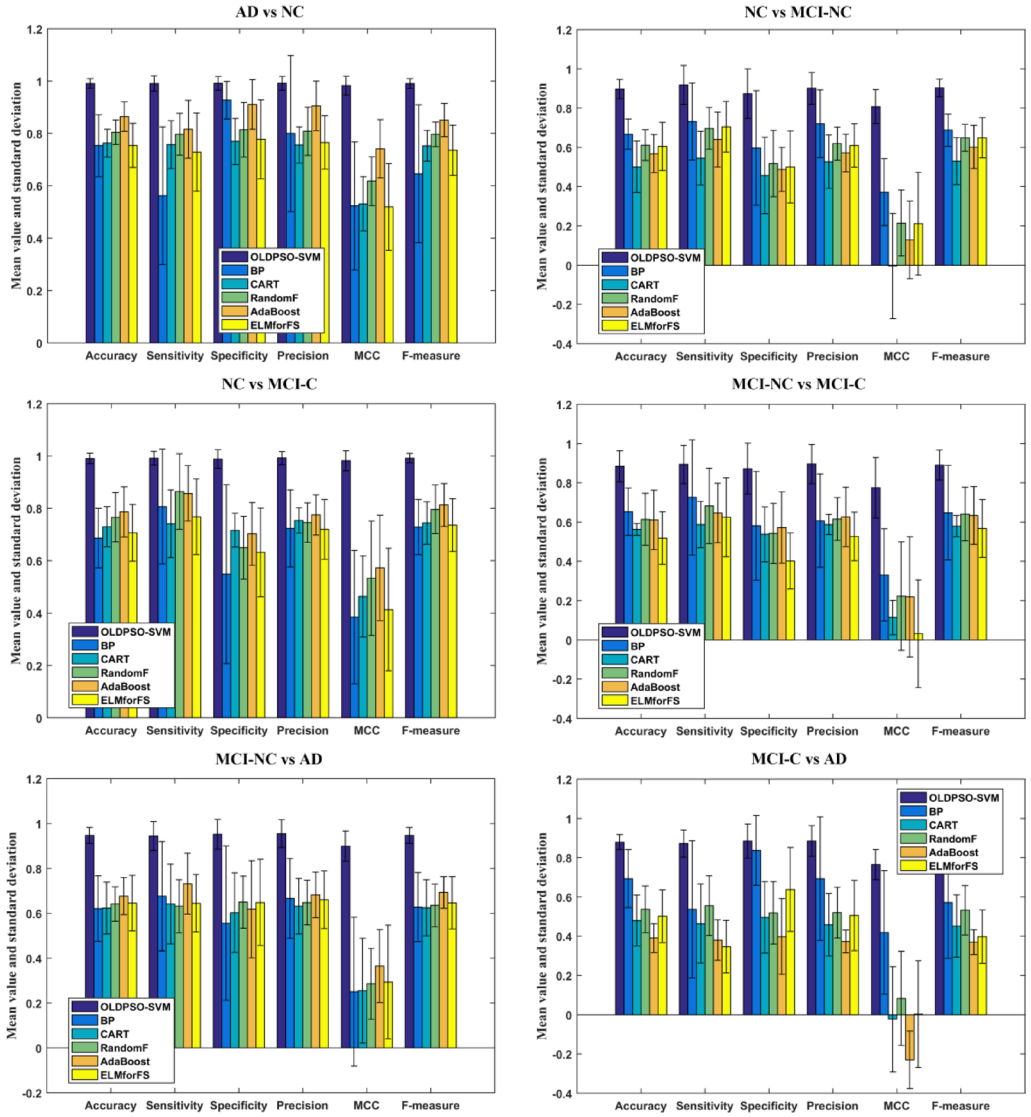


Fig. 17. Performance comparison of OLDPSO-SVM and baseline models for AD classification.

gyrus and the middle temporal gyrus were selected, highlighting the involvement of auditory and language processing areas in MCI progression. These regions are often associated with higher-order cognitive functions, suggesting that subtle changes in these areas could predict the transition from non-converting MCI (MCI-NC) to converting MCI (MCI-C). Similarly, the selected WM regions, including the putamen and posterior cingulate gyrus, emphasize the importance of these regions in cognitive control and memory networks, further underscoring their role in the conversion process.

These findings are consistent with previous research emphasizing the hippocampus, temporal gyrus, and parahippocampal regions as critical for the diagnosis of AD and MCI. However, this study extends the current understanding by integrating both GM and WM features, highlighting the significance of WM abnormalities in the disease progression. While GM atrophy has long been recognized as a biomarker for AD, the inclusion of WM regions in this analysis provides a more comprehensive view of the structural changes associated with both AD and MCI. Specifically, the frequent selection of WM regions, such as the parahippocampal gyrus and insula, underscores their pivotal role in the disease classification process. Overall, these feature selection results provide promising evidence for the integration of both GM and WM features in diagnostic models for AD and MCI. The regions identified in this study not only reinforce known disease mechanisms but also suggest the importance of lesser-studied regions, such as the olfactory and auditory areas, in early-stage detection. These findings could guide future research in developing more accurate multimodal diagnostic tools that combine both GM and WM markers to distinguish between NC, MCI, and AD with greater precision.

Despite the superior performance of the proposed OLDPSO-SVM model in feature selection and AD classification, certain limitations remain. First, this study is conducted using the ADNI dataset, which, although widely used in AD research, has inherent constraints in sample size and population diversity, potentially affecting the model's generalizability. Future studies should validate the robustness of the proposed method

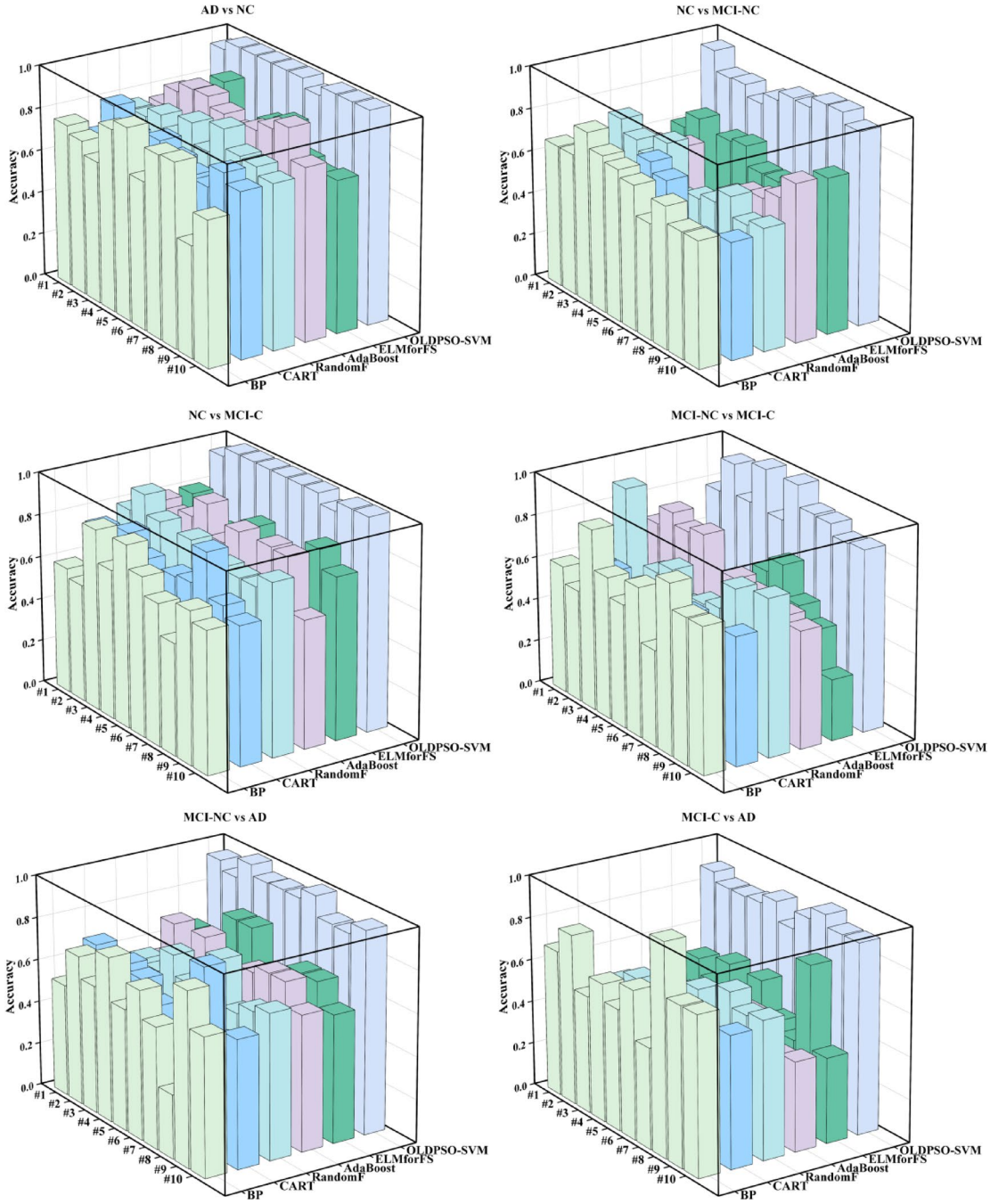


Fig. 18. Accuracy performance of various machine learning models across 10-fold cross-validation for different AD classification tasks.

on larger and more diverse datasets. Second, this study primarily utilizes static structural MRI data without incorporating functional MRI. Integrating multimodal imaging data in future research could further enhance diagnostic accuracy and stability. Additionally, longitudinal data analysis would be beneficial for assessing the predictive capability of the selected features in monitoring the progression from MCI to AD. Moreover, there is still room for further research regarding parameter selection. Therefore, as part of our future research endeavors, we will conduct systematic experiments to analyze the impact of different parameters (such as population size and learning factors) on algorithm performance to identify more robust parameter settings. Additionally, a more comprehensive comparison of the algorithm's efficiency, particularly in terms of memory usage and scalability for larger problems, is needed. To better assess the balance between efficiency and performance, we plan to explore these aspects in future work, incorporating further analyses to illustrate the tradeoffs between performance and computational cost when compared to existing algorithms. Finally, this study primarily focuses on the classification of AD and its early stage, MCI, without evaluating the applicability of the proposed method to other neurodegenerative diseases. Given that machine learning and deep learning have been successfully

References	Methods	Accuracy
Payan et al.[47]	CNN	95.39%
Billones et al.[48]	CNN	98.33%
Choi and Jin[49]	CNN	96%
Yagis et al.[50]	CNN	73.4%
Tufail et al.[51]	CNN	89.21%
Lyu et al.[52]	Transformer	96.8%
Jang and Hwang[53]	Transformer	93.21%
Li et al.[54]	Transformer	93.9%
Kushol et al.[55]	Transformer	88.2%
Liu et al.[56]	Transformer	99.4%
Chen et al.[57]	CNN + Transformer	97.65%
This study	Proposed OLDPSO-SVM	99.111%

Table 13. Comparison of OLDPSO-SVM with other existing methods for AD vs. NC.

applied to various medical imaging and health assessment tasks, including osteoporosis detection[58], tumor classification[59], AI-driven health monitoring fusion[60], and reinforcement learning strategies in medical data analysis[61], future research should explore its generalizability to a broader range of disease classifications.

Significance statement

As the global population continues to age, Alzheimer's Disease (AD) has emerged as a major public health concern. Developing effective early diagnostic methods for AD is crucial for improving patient outcomes and delaying disease progression. An enhanced Particle Swarm Optimization (PSO) algorithm, termed OLDPSO, is proposed to improve feature selection and classification. OLDPSO integrates opposition-based Latin squares sampling initialization with dynamic inertia weights and learning factors, overcoming traditional PSO limitations. Combining GM and WM features yields higher diagnostic accuracy than using either alone and identifies key brain regions associated with AD progression. The OLDPSO-SVM model can enhance diagnostic performance and provides valuable insights for developing MRI-based multimodal diagnostic tools for AD.

Conclusions

In this study, we developed an enhanced PSO algorithm, termed OLDPSO, to improve feature selection and classification accuracy within an SVM model for AD diagnosis using MRI-derived GM and WM volumes. MRI provides a non-invasive means to observe structural changes in the brain, where GM and WM volumes serve as significant biomarkers for AD. Traditional PSO algorithms face limitations in optimizing complex high-dimensional data such as GM and WM volumes. To address these challenges, OLDPSO integrates opposition-based Latin squares sampling for initialization with dynamic adjustments to inertia weights and learning factors. Benchmark experiments demonstrated that OLDPSO outperforms existing PSO variants in both solution quality and convergence speed, underscoring its efficiency in feature selection.

Validation using data from the ADNI confirmed the OLDPSO-SVM model's superior performance in distinguishing AD, MCI, and NC groups, with notable accuracy in classifying MCI subtypes (MCI-NC and MCI-C). Furthermore, the model's diagnostic accuracy was enhanced by identifying optimal TFs for each classification task. The results also revealed that combining GM and WM volumes yields higher classification accuracy than using either feature alone and identifying key brain regions associated with AD progression.

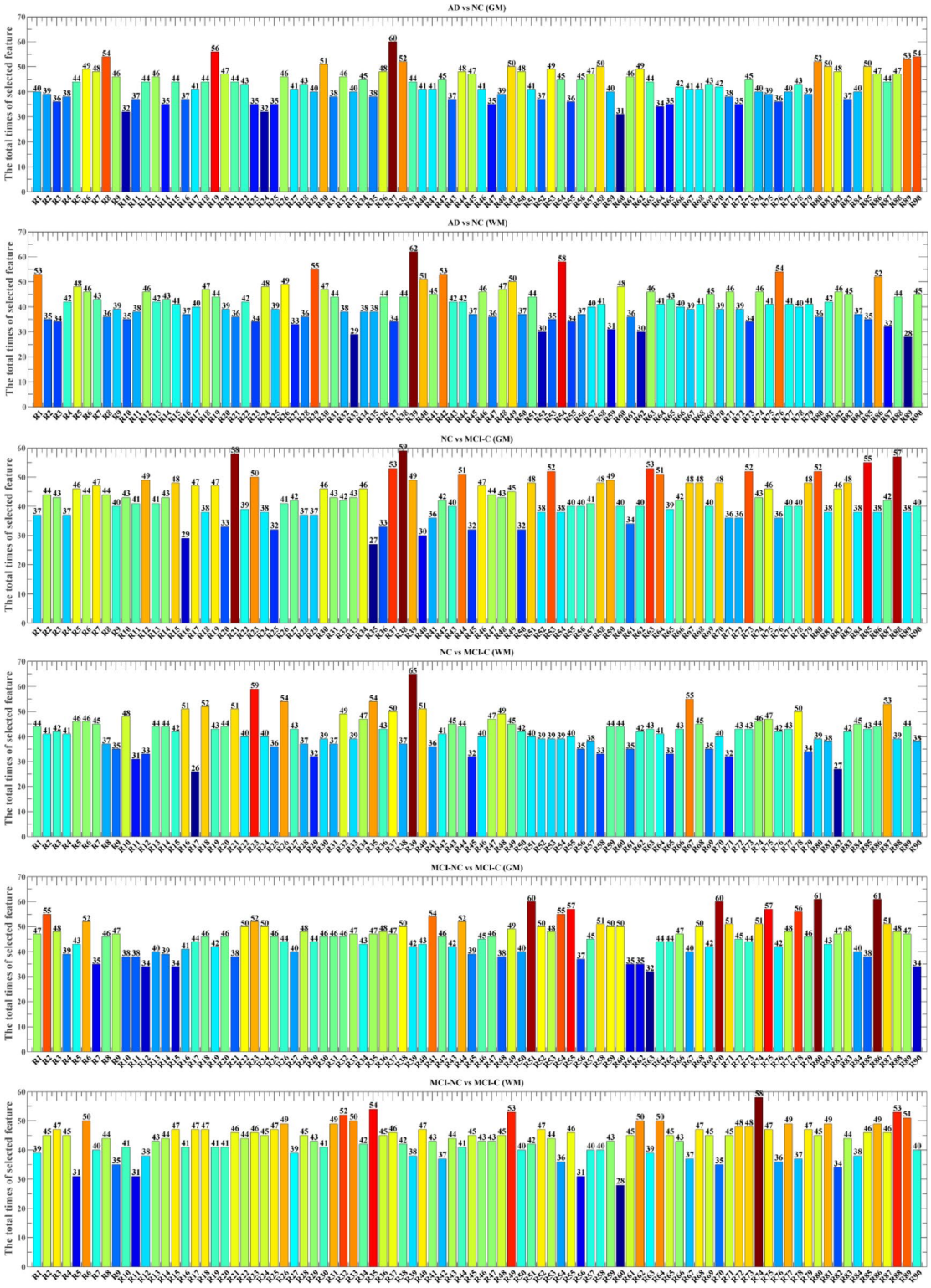


Fig. 19. Comparison of GM and WM feature selection in AD, MCI, and NC.

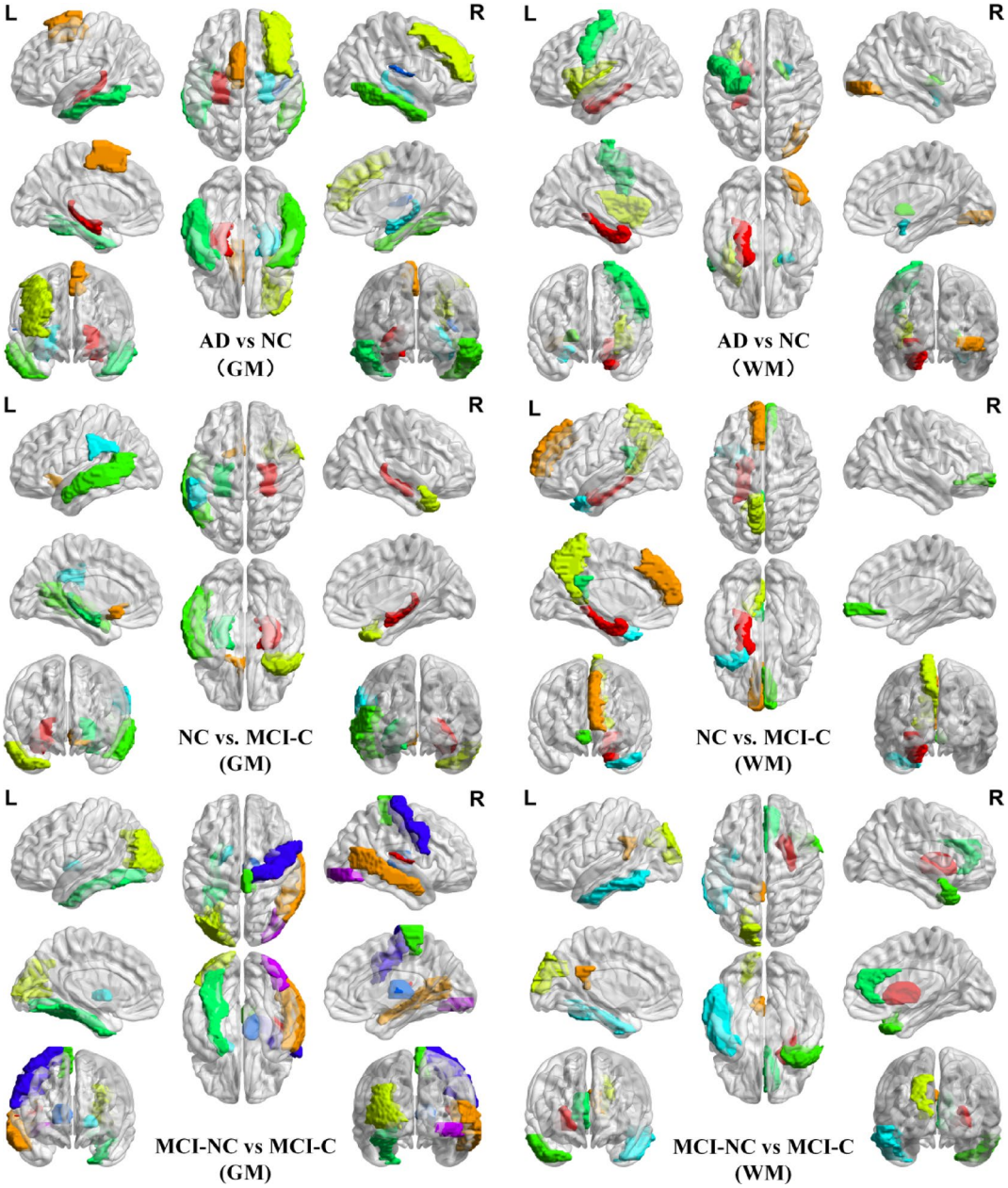


Fig. 20. Comparative analysis of structural differences in GM and WM among AD, MCI, and NC. The structural differences were visualized using BrainNet Viewer with version 1.7 (http://www.nitrc.org/projects/bnv/).

Data availability

The data supporting this research is derived from the Alzheimer's Disease Neuroimaging Initiative (ADNI) and can be accessed from the ADNI database (adni.loni.usc.edu) upon registration and adherence to the data usage agreement. Additional information can be provided by corresponding authors upon reasonable request.

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Author contributions

Jinhua Sheng designed the project and supervised the overall research; Qian Zhang collected data and analysis, performed experiments; Rougang Zhou, Qiao Zhang, Binbing Wang, and Rong Zhang participated in data collection and analysis. Qian Zhang and Jinhua Sheng wrote the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Ethical approval

This study was approved by the institutional review board (IRB) at Hangzhou Dianzi University (IRB-2020001) and the ethics committee at Beijing Hospital (2022BJYYEC-375-01).

Additional information

Correspondence and requests for materials should be addressed to J.S.

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